Measurement of the Second Knee in the Cosmic Ray Spectrum with the Surface Detector of the Pierre Auger Observatory

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and the

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of

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from Ciudad Autónoma de Buenos Aires

Day of the oral examination:

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Co-referee: Prof. Dr. Ralph Engel

Supervisor: Dr. David Schmidt

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Tag der mündlichen Prüfung:

Referent: Prof. Dr. Diego Ravignani Guerrero

Korreferent: Prof. Dr. Ralph Engel

Betreuer: Dr. David Schmidt

Abstract

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Chapter 1

The SD-433 dataset

The measurement of the cosmic-ray energy spectrum poses several experimental challenges. Since this observable carries important astrophysical information, the construction of a clean and well-understood dataset is essential. This requires not only a reliable event reconstruction, but also a careful characterization of the detector response, the trigger hierarchy, and the stability of the array over time.

This chapter introduces the main elements that define the dataset used in this work. The SD-433 is a compact array of 19 surface-detector stations arranged on a triangular grid with a spacing of 433 m, nested within the SD-750 array to extend the energy-spectrum measurement toward lower energies [1]. It begins with the hierarchical trigger chain, from the local station triggers to the selection of physical events, described in Section 1.1. Section 1.2 addresses the trigger efficiency and establishes the energy threshold above which the exposure can be treated as purely geometrical. The discussion closes with the stability and operational history of the array in Section 1.3, which provides the basis for the exposure calculation in Section 1.4 used in the spectrum measurement.

1.1 The Surface Detector Triggers

To discriminate signals produced by an Extensive Air Shower (EAS), initiated by the interaction of a primary cosmic ray (CR) in the atmosphere, the surface detector uses a hierarchical trigger system. This hierarchy begins at the level of individual stations and continues up to the selection of physical events and quality requirements at the array level. A schematic summary of the trigger chain is shown in Fig. 1.1.

Originally, two T1 trigger modes were implemented at the station level: simple treshold (Th)-T1, and time over threshold (ToT). The Th-T1 trigger requires a signal larger than $1.75\ I_{\mathrm{VEM}}^{\mathrm{peak}}$ in each of the three photomultiplier tube (PMT)s. This trigger is designed to select large signals that are not necessarily spread in time and is particularly sensitive to inclined showers with a stronger muonic component.

The ToT trigger requires that at least 13 bins within a sliding window of 3 µs exceed $0.2\ I_{\mathrm{VEM}}^{\mathrm{peak}}$ in coincidence in at least two PMTs. This trigger is designed to select sequences of smaller signals spread over time, as expected from the electromagnetic component of the shower. It is therefore well suited to lower-energy showers and to events in which the detector is far from the shower core.

In 2014, two additional station-level trigger modes were introduced. These triggers were designed to improve sensitivity to the electromagnetic component of the shower and to extend the energy range over which the array can detect events.

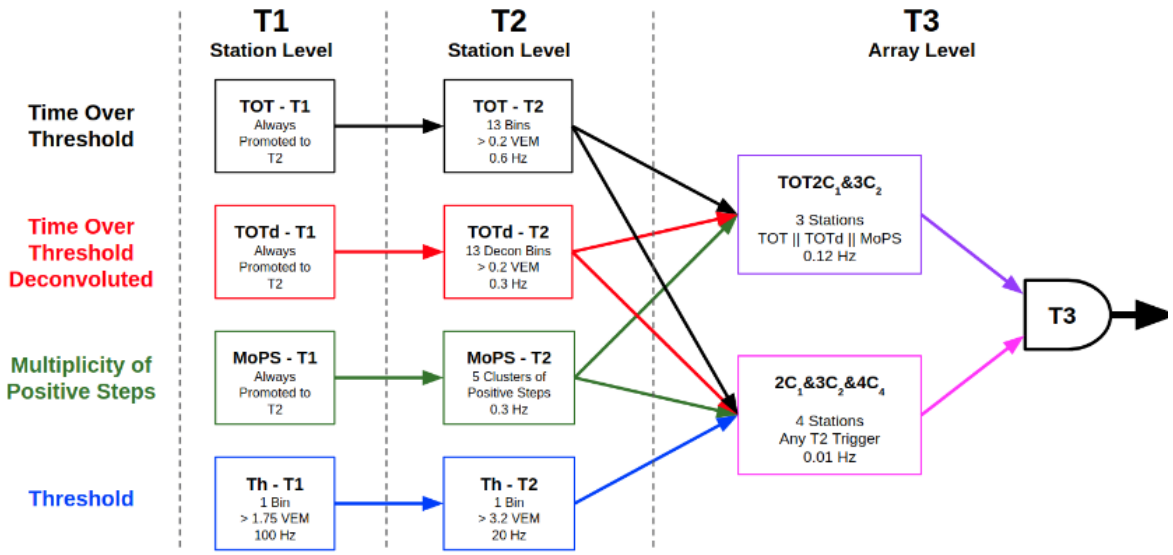


Figure 1.1: Summary of the T1, T2, and T3 trigger levels, their requirements, and their typical rates. The ToT, ToTd, and MoPS triggers are automatically promoted to the T2 level.

The first of these triggers is the Time-over-Threshold-deconvolved (ToTd) trigger. Traces with a significant electromagnetic component resemble a ToT signal, with a fast rise followed by an exponential decay. After deconvolution, the expected signal becomes approximately flat, while preserving the fluctuations associated with particle counting. The ToT requirement is then applied to the deconvolved trace.

The second additional trigger is the Multiplicity-of-Positive Steps (MoPS) trigger. This trigger is independent of the Vertical Equivalent Muon (VEM) calibration and is based on the signal pattern expected from multiple low-amplitude particles arriving over an extended time interval. It counts the number of positive steps within a 3 µs sliding window, where a positive step is defined as an increase in signal amplitude in consecutive bins. To be counted, the step must be larger than the typical noise level and smaller than the amplitude expected from a vertical muon. If more than four such positive-step instances are found, a MoPS trigger is issued.

The second trigger level, T2, is applied locally at each station to reduce the trigger rate to a value compatible with the bandwidth of the communication system. All ToT, ToTd, and MoPS triggers are automatically promoted to T2. By contrast, a Th-T1 trigger is promoted to T2 only if the signal exceeds a higher threshold of 3.2 $I_{\mathrm{VEM}}^{\mathrm{peak}}$ in coincidence in the three PMTs. At this stage, the trigger rate is reduced to about 20 Hz per station.

The T2 information, namely the trigger type and time stamp, is then sent to the Central Data Acquisition System (CDAS), where array-level triggers are formed from spatial and temporal coincidences among stations. This defines the third trigger level, T3, which initiates the data acquisition for the event. Once a T3 is issued, the flash analog-to-digital converter (FADC) traces from all stations with a T2 trigger are sent to CDAS, together with those from stations that raised a T1, provided that their trigger time lies within 30 µs of the T3 timestamp.

The search for temporal and spatial groupings is conducted using two types of T3 algorithms. Both of them look for T2-triggered stations in a compact configuration. The compactness is described by the number of crowns over which the triggered stations are spread. A crown is a way of counting hexagons outward from a central station.

The first configuration, *ToT2C1&3C2*, requires a coincidence of at least three nearby stations passing any combination of ToT, ToTd, and MoPS triggers. These stations must be

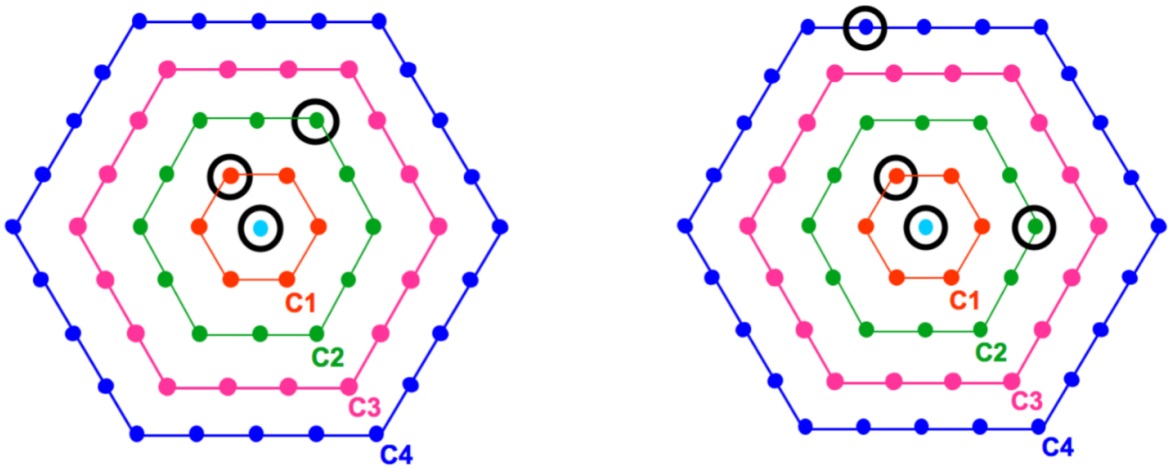


Figure 1.2: Examples of the two T3 triggering modes. Left: the three-fold configuration *ToT2C1&3C2*. Right: the four-fold configuration *2C1&3C2&4C4*. Figure adapted from Fig. 2.14 of Ref. [2].

in a compact pattern, such that one station has another one in the first crown and a third one in the second crown. Since the ToT trigger has a very low background rate, this T3 mode has a low contamination and is highly efficient for showers with zenith angle below 60°.

The second T3 configuration, denoted *2C1&3C2&4C4*, is more permissive. It requires a four-fold coincidence of T2 stations, independently of their trigger type. In this case, one triggered station must have another in the first crown, a third one in the second crown, and the fourth may be as far as the fourth crown. This configuration is particularly relevant for more inclined showers, whose triggered stations can be spread over a wider area on the ground.

The T1, T2, and T3 selection are performed online and the data is stored to be reconstructed later, where the next two trigger levels are applied. The physical trigger, T4,is designed to reject accidental coincidences and retain physically originated showers. For vertical showers, two T4 criteria are used.

The first T4 criterion, denoted *3ToT*, originally required three nearby stations having a ToT trigger in a non-aligned pattern, but was later updated to also include combinations of the ToTd and MoPS triggers. In addition, the trigger times of the three stations must be compatible with a plane shower front moving at the speed of light.

The second T4 criterion, denoted *4C1*, requires four nearby triggered stations without imposing any condition on the trigger type. As in the 3ToT case, their times must be compatible with a plane shower front moving at the speed of light. This criterion complements the 3ToT selection and improves the efficiency for events that are not selected by the more restrictive ToT-based condition.

To ensure a reliable reconstruction of the event, the 6T5 fiducial quality trigger is applied. It requires the station with the highest signal to be surrounded by six active neighbouring stations at the time of the event. This condition ensures that the shower core is well contained within the array, improves the accuracy of the reconstructed quantities, and allows the effective area to coincide with the geometrical one for energies higher than full efficiency.

The 6T5 trigger also defines a fiducial area associated with each active station. Because the surface-detector stations are arranged on a triangular grid, this area is a regular hexagon centred on the station with the largest signal. Requiring all six neighbouring stations to be active ensures that this station does not lie at the boundary of the active array and provides

adequate containment of the reconstructed shower core. For a station spacing $d = 433\,\mathrm{m}$, the area of this hexagon is

$$S_{\mathrm{hex}} = \frac{\sqrt{3}}{2} d^2 = 0.162\,\mathrm{km}^2. \tag{1.1}$$

The event sample used in this work therefore consists of physical events selected by the T4 trigger that also satisfy the 6T5 fiducial requirement.

1.2 Trigger Efficiency

In order to calculate the exposure from the purely geometrical aperture of the array, it is necessary to establish an energy threshold above which the detector can be treated as fully efficient. Below this threshold, the probability of detecting and reconstructing a shower depends on the primary energy and arrival direction, and the exposure can no longer be calculated from the geometry alone. The spectrum measurement is therefore restricted to an energy range in which the array can be treated as fully efficient [3]. The efficiency parameterization is nevertheless included in the detector response used in the forward-folding procedure to account for a residual detector influence near the threshold.

The efficiency of the SD-433 array was studied in detail in Ref. [4], using simulated showers and validating the results with measured data. For this purpose, CORSIKA simulations were performed with EPOS-LHC [5] as the high-energy hadronic-interaction model and FLUKA [6] at lower energies. Seven primary mass groups, ranging from proton to iron, were simulated and reconstructed using the SD-433 array. Additionally, a mixed-composition result was estimated using the Global Spline Fit (GSF) mix [7]. The study considers the probability that a shower with a given energy and zenith angle is successfully reconstructed by the array, after requiring the 6T5 condition and including the contribution of the ToTd and MoPS station triggers. The efficiency is defined as the ratio of the number of simulated events successfully reconstructed by the SD-433 to the total number of simulated events in the corresponding energy interval. The simulation-based efficiency is transformed to the SD-433 energy scale using the attenuation and calibration chain adopted for data [4].

Efficiency studies of the SD-433 are customarily restricted to events with $\theta < 45^\circ$, since the quality of the standard geometry reconstruction degrades at larger zenith angles [4]. In this work, the more conservative selection $\theta < 40^\circ$ is adopted to match the zenith-angle range used for the SD-750 [8].

The resulting efficiency in the SD-433 energy scale is determined with an unbinned likelihood fit, considering if each simulated shower triggers the array or not. The corresponding efficiency estimators and fits are shown in Fig. 1.3 for the GSF mix, together with proton and iron as extreme cases. For events with zenith angles $\theta < 40^\circ$, the efficiency for the GSF mix is parameterized as

$$\epsilon(E) = \Phi\left(a_0 + a_1 \lg\left(\frac{E}{10^{17}\,\mathrm{eV}}\right)\right), \tag{1.2}$$

where Φ is the cumulative distribution function of the standard normal distribution, E is the primary energy in the SD-433 energy scale, a_0 determines the energy of the efficiency transition, and a_1 governs its steepness. The fitted parameters for the GSF mix are $a_0 = 4.21 \pm 0.03$ and $a_1 = 5.50 \pm 0.04$. This parameterization will later be used in the forward-folding procedure described in Chapter 3.

Below full efficiency, the efficiency for proton showers is higher than that for iron showers. This difference is linked to the deeper development of lighter-primary showers combined

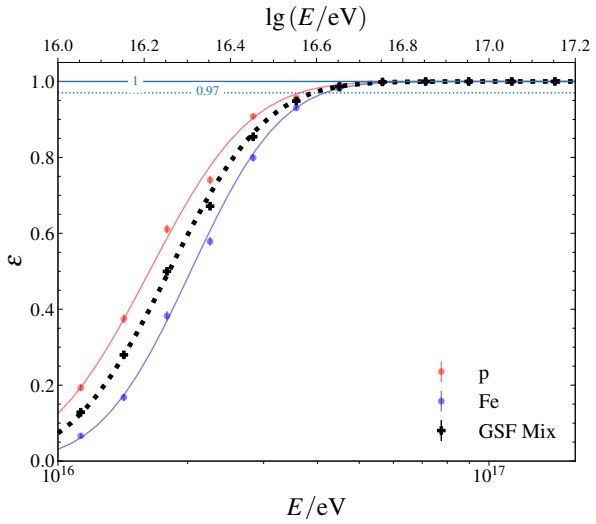


Figure 1.3: Efficiency curves as a function of the logarithm of the SD-433 energy. Markers correspond to binned estimates of $\epsilon(E)$ obtained by dividing the number of triggered simulated showers by the total number of triggered and non-triggered simulated showers, while the lines show the unbinned fits. The inset highlights the region near full efficiency. Figure adapted from Fig. 7 of Ref. [4].

with the enhanced sensitivity of the second-generation triggers to the electromagnetic component of the shower [4]. For proton, iron, and the GSF mix, the first energy bin with an efficiency above 97% begins at $\lg(E/\mathrm{eV}) = 16.7$. This simulation-based result motivates the data-driven validations described below, which support the more conservative threshold of $\lg(E/\mathrm{eV}) = 16.8$ adopted for the spectrum measurement.

The simulation-based efficiency result was cross-checked using two tests applied to reconstructed data. These tests are dependent of the core and energy reconstruction procedures, since both require reconstructed event observables and calibrated energies. Nevertheless, they probe two distinct signatures expected above full efficiency: the uniformity of the arrival directions in $\sin^2\theta$ and the uniform distribution of reconstructed shower cores within the fiducial 6T5 area. Together, they provide data-driven validation of the threshold inferred from simulations.

The first data-driven validation tests the isotropy of the arrival directions above different energy thresholds, following Ref. [4]. For isotropic arrival directions, the distribution in $\sin^2\theta$ is expected to be uniform up to the maximum zenith angle considered. For each energy threshold, the cumulative distribution of $\sin^2\theta$ was compared with the uniform expectation using the Anderson–Darling test statistic, which gives greater weight to deviations near the tails of the distribution. The reference distribution of the test statistic was generated from pseudo-experiments uniformly distributed in $\sin^2\theta$, and the p-value was calculated as the fraction of simulated statistics larger than the value observed in data.

The test gives numerical p-values of 0, 0.008, and 0.357 for minimum-energy thresholds of $\lg(E/\mathrm{eV}) = 16.7$, 16.8, and 16.9, respectively. Using $p < 0.005$ as the rejection criterion, the isotropic hypothesis cannot be rejected for events with $\lg(E/\mathrm{eV}) \geq 16.8$. This supports the choice of $\lg(E/\mathrm{eV}) = 16.8$ as a conservative threshold above which the array can be treated as fully efficient.

As a second validation, the spatial distribution of the reconstructed shower cores was examined using events with $\theta < 40^\circ$. Once the detector is fully efficient, an isotropic flux is expected to produce a uniform distribution of shower cores within the fiducial 6T5 area. To combine the equivalent regions associated with different stations, the position of the station

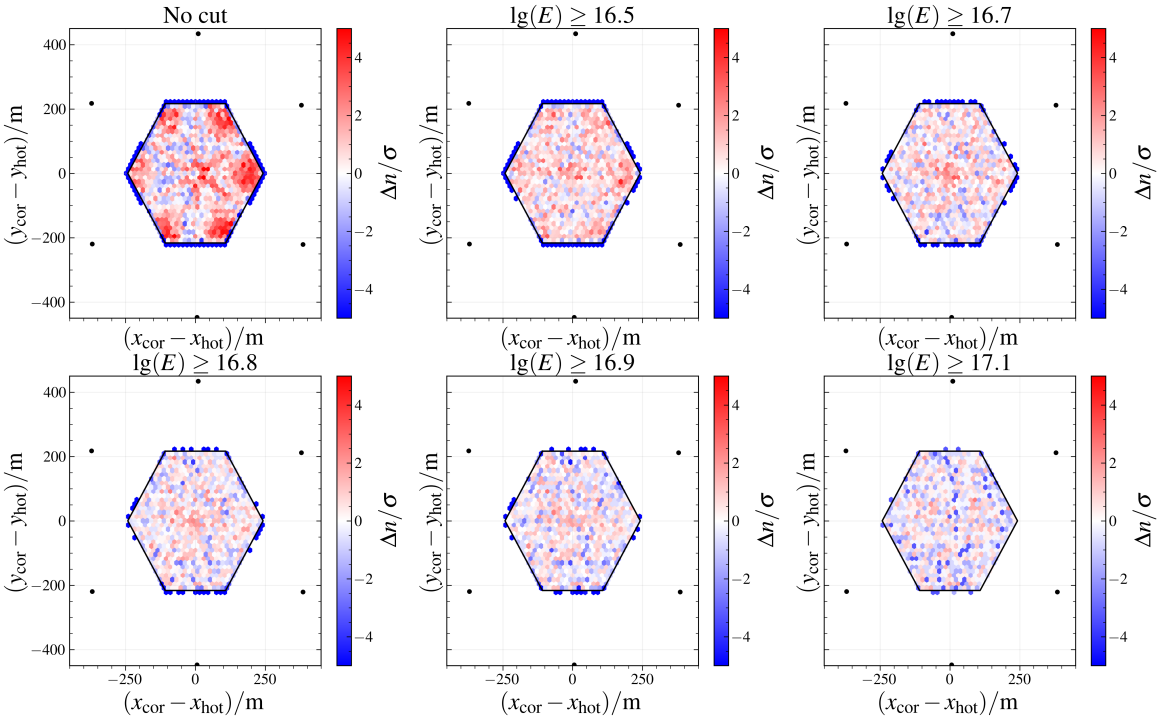


Figure 1.4: Distribution of reconstructed core positions in the fiducial area for different minimum-energy thresholds, using events with $\theta < 40°$. The outermost bins are excluded to avoid edge effects associated with events reconstructed close to the fiducial boundary. At $\lg(E/\mathrm{eV}) = 16.7$, an overdensity is still visible around the central station, consistent with the reconstruction pulling the core toward the hottest station. From $\lg(E/\mathrm{eV}) = 16.8$ onward, the distribution becomes more uniform.

with the largest signal is subtracted from the reconstructed core position, folding all events into a single hexagon.

The resulting core positions are grouped using hexagonal bins with a characteristic size of approximately 28 m. This choice preserves the symmetry of the detector and gives a spatial scale comparable to the core-position resolution, which is approximately 20 m in this energy range [9]. Bins adjacent to the boundary of the fiducial hexagon are excluded because fluctuations in the reconstructed core position can cause events near the edge to migrate across the boundary and distort the measured occupancy, causing border effects.

Figure 1.4 shows the standardized deviation of the occupancy of each accepted bin from the mean occupancy,

$$\Delta n_i = \frac{n_i - \bar{n}}{\sigma_n}, \tag{1.3}$$

where n_i is the number of reconstructed cores in bin i, $\bar{n}$ is the mean occupancy of the accepted bins, and σ_n is the standard deviation of their occupancies.

Without a minimum-energy requirement, the maps exhibit localized overdensities associated with reconstruction artifacts below full efficiency. These structures become less pronounced as the minimum energy increases and the core reconstruction improves. At $\lg(E/\mathrm{eV}) = 16.7$, an overdensity remains around the central station, consistent with reconstructed core positions being preferentially drawn toward the station with the largest signal. This feature is no longer apparent from $\lg(E/\mathrm{eV}) = 16.8$ onward, for which the cores populate the fiducial area more uniformly.

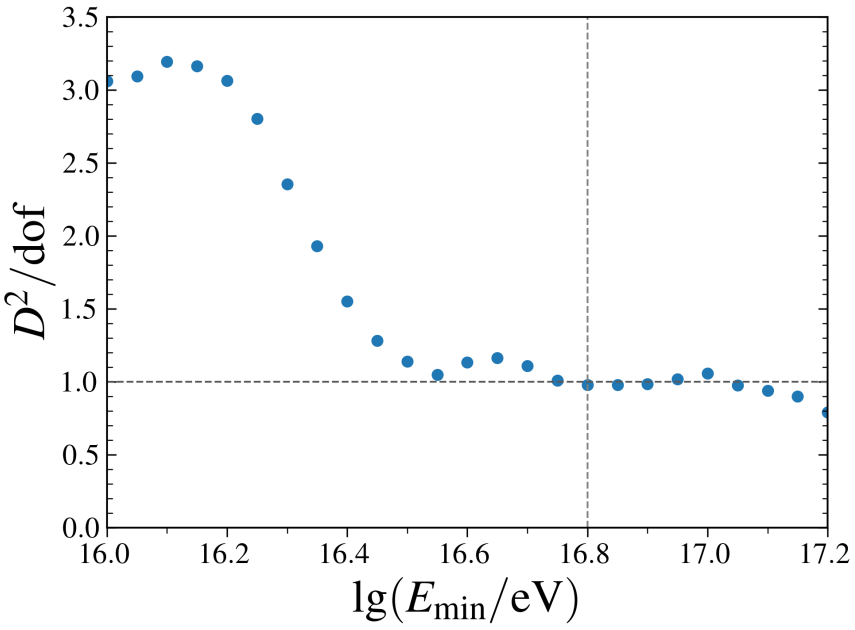


Figure 1.5: Reduced Poisson deviance of the reconstructed core distribution with respect to a uniform expectation as a function of the minimum-energy threshold, for events with $\theta < 40^\circ$. The reduced deviance falls below unity at $\lg(E_{\mathrm{min}}/\mathrm{eV}) = 16.8$, supporting the adoption of this value as the beginning of the full-efficiency regime. Its increase at the highest thresholds is associated with the larger statistical fluctuations caused by the lower bin occupancies.

To quantify the uniformity of the core distribution, the observed bin occupancies were compared with a uniform expectation using the Poisson deviance. For a sample containing N events distributed among N_{bin} accepted bins, the expected occupancy under the uniform hypothesis is,

$$\nu = \frac{N}{N_{\mathrm{bin}}}. \tag{1.4}$$

The Poisson deviance is then

$$D^2 = 2\sum_i \left[\nu - n_i + n_i \ln\left(\frac{n_i}{\nu}\right)\right], \tag{1.5}$$

where n_i is the observed occupancy of bin i. Since the overall normalization is determined from the data, the number of degrees of freedom is $N_{\mathrm{bin}} - 1$. The reduced deviance, D^2/dof, is used to compare samples selected using different minimum-energy thresholds.

The resulting reduced deviance, D^2/dof, is shown in Fig. 1.5. It decreases as the minimum-energy threshold increases and falls below unity at $\lg(E_{\mathrm{min}}/\mathrm{eV}) = 16.8$. This result supports the use of $\lg(E/\mathrm{eV}) = 16.8$ as the lower boundary of the full-efficiency regime for events with $\theta < 40^\circ$.

The simulation-based efficiency determination and the two data-driven validations therefore support the adoption of $\lg(E/\mathrm{eV}) = 16.8$, corresponding to 0.063 EeV, as the full-efficiency threshold for the SD-433 dataset. Above this threshold, the exposure can be calculated from the geometric aperture of the array given that the efficiency factor in this calculation is unity

1.3 Array stability

The SD-433 evolved from an engineering array around the Tina Turner station into its seven-hexagon configuration over a six-year period. During this time, changes to the detector layout and communication system affected both the number of active stations and the information available to evaluate the 6T5 condition.

The main stages relevant to the dataset used in this work are summarized below [10]:

- April 2013: stable data acquisition began with a single hexagon.
- January 2014: the ToTd and MoPS triggers were deployed.
- June–December 2017: the stations Chichino Jr. and Guili Jr. were repositioned.
- January 2018: the array was expanded to four hexagons.
- July 2018: silent stations were included in the acquired events.
- May 2019: installation of the seven-hexagon configuration was completed.
- December 2021: the electronics upgrade for Phase II began.

The status of non-triggered stations was not included in the event records in the origina array. Consequently, the 6T5 trigger could not be evaluated, resulting in a substantial deficit of events satisfying this quality requirement. Although the earlier data were subsequently recovered using monitoring information [10], the dataset used in this work begins on 1 July 2018, when silent stations were included in the events. Restricting the analysis to this period ensures that the 6T5 trigger is evaluated directly from the detector-status information included in the acquired event records and provides a homogeneous dataset throughout the analysis. The selected period ends on 21 December 2021, when the electronics upgrade of the SD-433 array began.

The stability of the array is monitored using the daily event rate shown in Fig. 1.6.

The increase in the total event rate observed in 2019 results from the addition of three new hexagons to the array. To separate changes in the array size from variations in its operational stability, the event rate is normalized by the number of active hexagons at each time, as shown in Fig. 1.6. For a stable detector configuration, the normalized rate is expected to remain approximately constant, so significant deviations can be used to identify changes in the acquisition conditions.

The operational status of each station is recorded every second and indicate whether a station is capable of acquiring data in the so-called T2Life flags. A hexagon is considered active when its central station and all six stations in its first crown are tagged as alive. The instantaneous array configuration can therefore be evaluated with a time resolution of one second [11]. The distribution of the daily 6T5 event rate normalized by the number of active hexagons has a mean of 6.8 events per day per hexagon and a standard deviation of $\sigma_{N_{\mathrm{evt}}} = 1.8$.

Although the surface detector has a duty cycle close to 100%, communication and electronics failures can produce periods of unstable data acquisition. These periods are identified from unexpectedly long time intervals between successive 6T5 events. The probability of observing a time interval longer than T is,

$$P(t > T) = \exp(-F\,A\,T), \tag{1.6}$$

where F is the expected event rate per unit time and unit area and A is the average effective area of the active hexagons during the interval. An interval is classified as unstable when

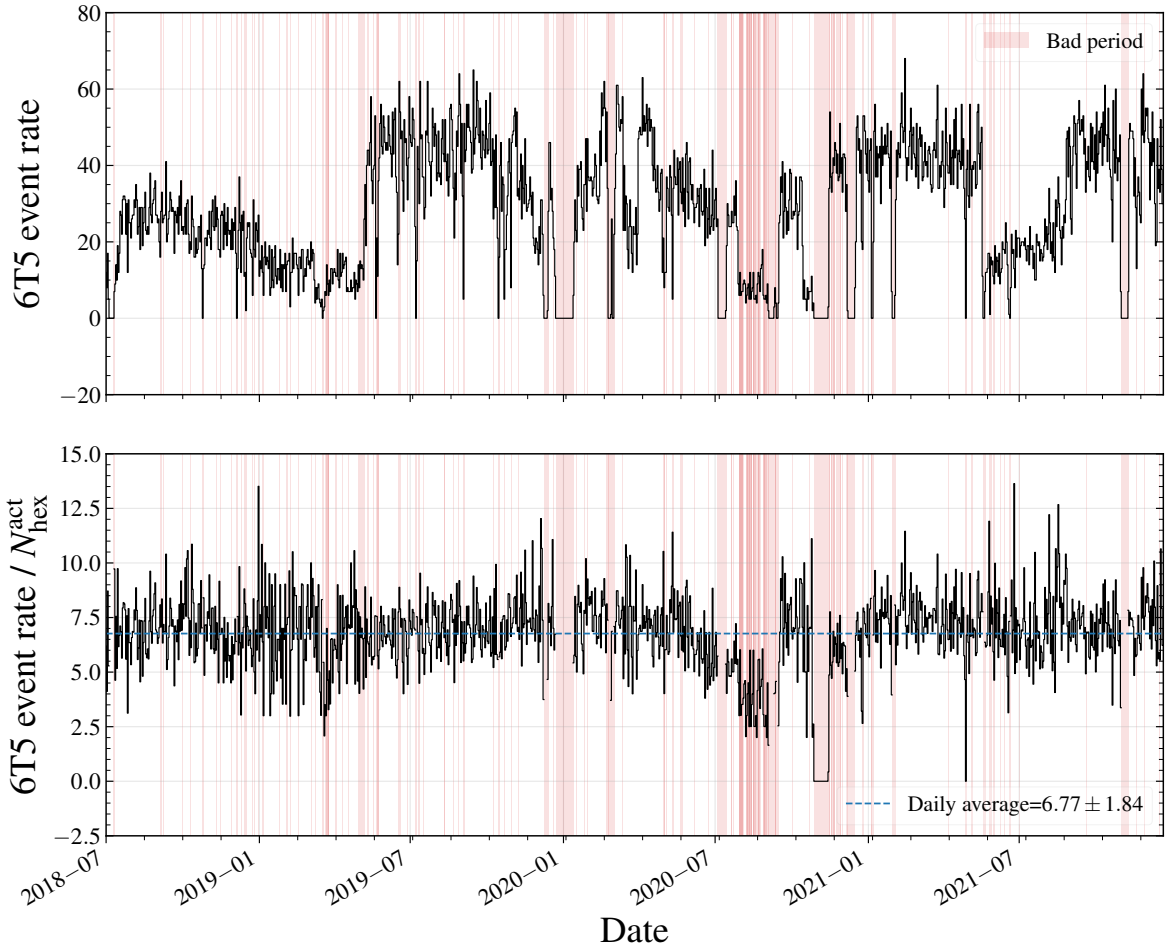


Figure 1.6: Top: Daily 6T5 event rate of the SD-433 array during the acquisition period used for the spectrum. Bottom: Daily event rate normalized by the number of active hexagons calculated from the T2Life information. The horizontal line indicates the average normalized event rate, and the shaded regions indicate the periods of unstable acquisition excluded from the analysis.

this probability falls below a threshold set to $\alpha = 10^{-2.5}$ [11]. The intervals satisfying $P(t > T) < \alpha$, referred to as *bad periods*, are shaded in Fig. 1.6. Together, these bad periods account for 170 days of the 1269-day acquisition interval, corresponding to an exposure loss of about 13%.

1.4 Exposure of the SD-433

After establishing the full-efficiency threshold and identifying the periods of stable acquisition, the exposure can be determined geometrically. The fiducial area associated with one active 6T5 hexagon is given by Eq. (1.1). Acceptance describes the ability of the array to collect particles from different directions. For a flat collecting surface, the area seen from an arrival direction is its projected area, $S_{\mathrm{hex}} \cos\theta$. The acceptance of an elemental hexagon in a acceptance cone delimited by the zenith angles θ_{min} and θ_{max} is found by integration that projected area over all accepted directions,

$$A_{6T5} = S_{hex} \int_0^{2\pi} d\phi \int_{\theta_{min}}^{\theta_{max}} \cos\theta \sin\theta \, d\theta \tag{1.7}$$

$$= \frac{\sqrt{3}}{2} d^2 \pi \left(\sin^2\theta_{max} - \sin^2\theta_{min} \right). \tag{1.8}$$

For the zenith-angle range used in this work, $\theta_{min} = 0^\circ$ and $\theta_{max} = 40^\circ$, the acceptance per active hexagon is $A_{6T5} = 0.211\,\mathrm{km}^2\,\mathrm{sr}$. The acceptance for the array containing N_{hex} active hexagons is,

$$\frac{d\mathcal{E}}{dt} = N_{hex}(t) A_{6T5}. \tag{1.9}$$

The accumulated exposure is obtained by integrating this acceptance over the selected acquisition intervals,

$$\mathcal{E} = \int_{\text{good periods}} N_{hex}(t)\, A_{6T5}\, dt, \tag{1.10}$$

where the identified bad periods are excluded.

The upper panel of Fig. 1.7 shows the daily mean number of active hexagons calculated from the minute-based T2Life information. Its evolution reflects both changes in the array configuration and temporary detector downtime. Their maximum corresponds to all 7 hexagons of the SD-433 in good working order. The lower panel shows the accumulated exposure contributed by each hexagon and their sum after excluding the bad-acquisition periods.

The total accumulated exposure for $\theta < 40^\circ$ is $2.85\,\mathrm{km}^2\,\mathrm{sr\,yr}$.

Applying the 6T5 quality condition, the full-efficiency threshold $\lg(E/\mathrm{eV}) \geq 16.8$, and the zenith-angle selection $\theta < 40^\circ$ yields 34 406events.

1.5 Chapter summary

This chapter presented the trigger, efficiency, and stability requirements used to define the SD-433 dataset for the measurement of the cosmic-ray energy spectrum. The hierarchical trigger system identifies physical air-shower events, while the 6T5 quality condition selects showers whose cores lie within an active region of the array and can therefore be reliably reconstructed.

The detector efficiency was determined using simulations and validated with measured data through the distributions of the arrival directions and reconstructed shower cores. These studies support the choice of $\lg(E/\mathrm{eV}) = 16.8$ as the lower boundary of the full-efficiency regime for events with $\theta < 40^\circ$. Above this threshold, the aperture can be calculated geometrically because the detection probability can be considered independent of the primary energy and mass.

The operational history and active configuration of the array were characterized using the T2Life information. Periods of unstable acquisition were identified and excluded, while the number of active 6T5 hexagons was used to integrate the time-dependent acceptance. The final dataset covers the period from 1 July 2018 to 21 December 2021 and has a live-time fraction of 87%. After applying the 6T5 quality condition, the full-efficiency threshold, and the zenith-angle selection, it contains 34 406events with an accumulated exposure of $2.85\,\mathrm{km}^2\,\mathrm{sr\,yr}$. This dataset is used in the following chapter to determine the SD-433 cosmic-ray energy spectrum.

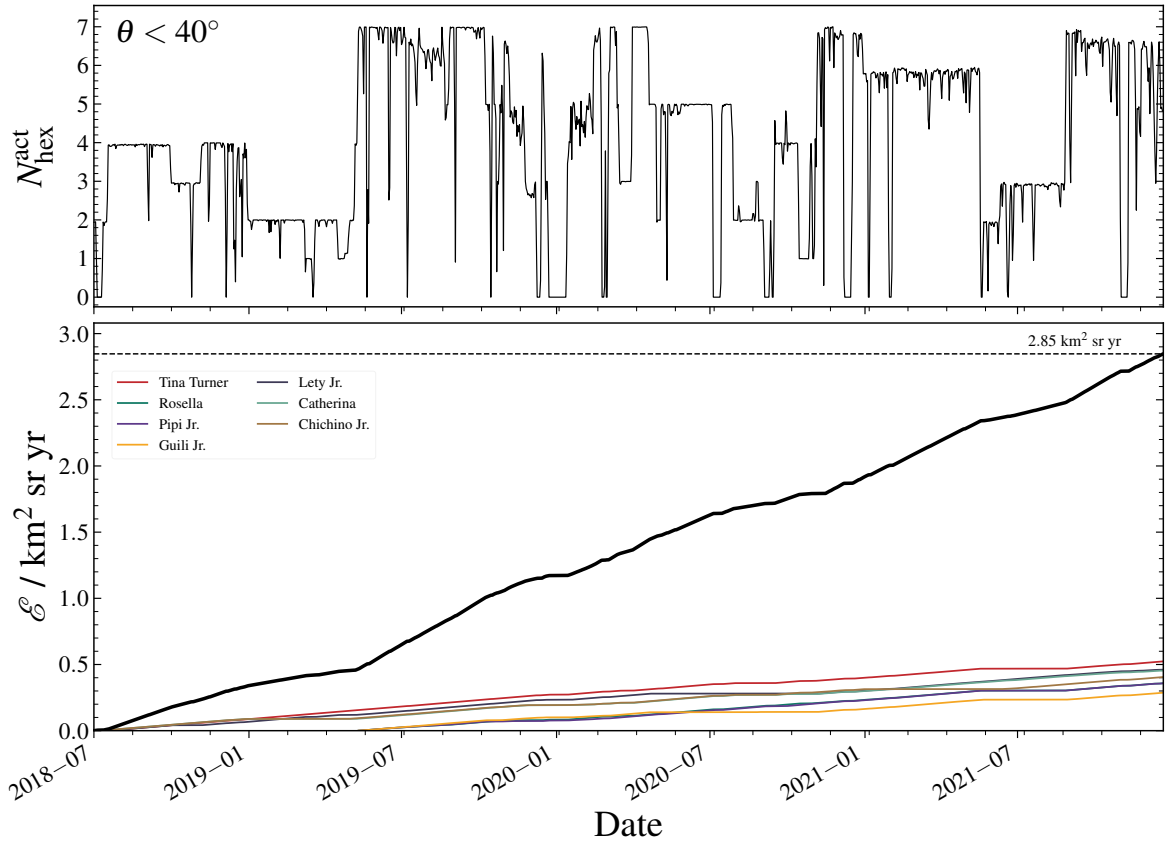


Figure 1.7: Evolution of the SD-433 array and its exposure during the selected data period. Top: daily mean number of active 6T5 hexagons calculated from the T2Life information. Bottom: accumulated exposure of the individual hexagons and their sum for $\theta < 40°$ after excluding bad-acquisition periods.

Chapter 2

Energy reconstruction

This chapter describes the reconstruction chain that connects the station-level measurements of an event to an estimate of the primary cosmic-ray energy. In Section 2.1, the Offline reconstruction is used to determine the shower geometry and to estimate the signal at a reference distance from the shower axis through a fit of the lateral distribution function (LDF). The zenith-angle dependence caused by atmospheric attenuation is then determined and removed in Section 2.2. Finally, in Section 2.3, the corrected shower size is calibrated against the energy estimator of the SD-750 to assign an energy to each SD-433 event.

2.1 Reconstruction of the shower size

The shower reconstruction proceeds in two stages. First, the relative start times of the signals measured by the participating stations are used to determine the arrival direction of the shower and the impact position of its core on the ground. The recorded signals are then fitted as a function of the perpendicular distance to the shower axis using an empirical LDF to model the decrease in the signal with the distance to the core. Analogously to the other SD arrays, the value of this function at the reference distance of 300 m, denoted by $S(300)$, provides the ground-level shower-size estimator used for the SD-433. Because the core position affects the station distances entering the LDF, the geometry and shower size are finally determined in a combined fit initialized with the results of the separate reconstructions. A detailed description of the event-reconstruction procedure used for the surface detector can be found in Ref. [12]. Here, the main elements relevant to the reconstruction of the SD-433 events are summarized, with particular emphasis on the steps implemented for the present analysis.

2.1.1 Geometrical reconstruction

The geometrical reconstruction uses the positions and signal start times of the detectors participating in the event. Its purpose is to determine the position of the shower core on the ground, $\vec{x}_c$, and the unit vector along the shower axis, $\hat{a}$, whose components are the direction cosines from which the zenith angle, θ, and the azimuth angle, ϕ, are obtained.

The reconstruction begins with a plane-front fit to the station start times, which provides an initial estimate of the arrival direction. The signal-weighted barycentre of the participating stations is used as the initial estimate of the core position. These estimates are subsequently refined using a curved description of the shower front.

In the $\overline{\text{Off}}\underline{\text{line}}$ reconstruction, the most advanced particles of the shower, which define the extreme shower front, are approximated by a sphere expanding at the speed of light from a virtual origin $\vec{x}_0$ at time t_0. The predicted time at which this front reaches a position $\vec{x}$ is

$$c\, t_{\text{sh}}(\vec{x}) = c\, t_0 + |\vec{x} - \vec{x}_0|. \tag{2.1}$$

The fit minimizes the squared differences between the measured and predicted start times,

$$\chi^2 = \sum_i \frac{[t_i - t_{\text{sh}}(\vec{x}_i)]^2}{\sigma_{t_i}^2}. \tag{2.2}$$

where t_i and σ_{t_i} are the start time measured by station i and its uncertainty, respectively.

The core position is refined together with the lateral distribution of the station signals. Once $\vec{x}_{\text{c}}$ is determined, the reconstructed shower direction is given by

$$\hat{a} = \frac{\vec{x}_0 - \vec{x}_{\text{c}}}{|\vec{x}_0 - \vec{x}_{\text{c}}|}. \tag{2.3}$$

The radius of curvature at the core position is

$$R_0 = |\vec{x}_0 - \vec{x}_{\text{c}}|\,. \tag{2.4}$$

For events with at least five participating stations, the radius of curvature is determined from the timing fit. For events with lower station multiplicity, the number of degrees of freedom is insufficient to constrain the curvature, and R_0 is fixed according to a parametrization obtained from higher-multiplicity events.

The left panel of Fig. 2.1 illustrates the geometrical reconstruction of an SD-433 event. The marker size is proportional to the station signal, while the colour scale represents its start time, progressing from earlier signals in blue to later signals in green. The black line indicates the reconstructed arrival direction projected onto the ground, and the cross marks the reconstructed core position.

2.1.2 Lateral-distribution function

The shower size is then estimated by modeling the fall-off of the signal $S(r)$ with the distance r to the shower axis using a LDF [13]. Intrinsic shower-to-shower fluctuations produce variations in the longitudinal development, such that individual events may be sampled at different stages of their development. As a consequence, the LDF exhibits variations on an event-by-event basis, preventing the reconstruction of all the parameters in the shower. Instead, the reconstruction is based on an *average* LDF, $\langle S(r)\rangle$, and the expected signal at a fixed reference distance, r_{opt}, is adopted as the shower size. The optimal distance of 300 m for the SD-433 is determined by fitting the average LDF and identifying the distance at which the reconstructed shower size is minimally dependent on the LDF slope, following the method described in Ref. [14]. The dependence of the optimal distance on the assumed LDF shape, the primary energy, and the zenith angle remains at approximately the 5% level [12]. The average LDF is described using an empirically-chosen model based on a modified Nishimura–Kamata–Greisen (NKG) function,

$$\langle S(r)\rangle = S(r_{\text{opt}}) \left(\frac{r}{r_{\text{opt}}}\right)^{\beta} \left(\frac{r + r_{\text{scale}}}{r_{\text{opt}} + r_{\text{scale}}}\right)^{\beta}, \tag{2.5}$$

where β is the LDF slope and $r_{\text{scale}} = 700\,\text{m}$ is a scale parameter.

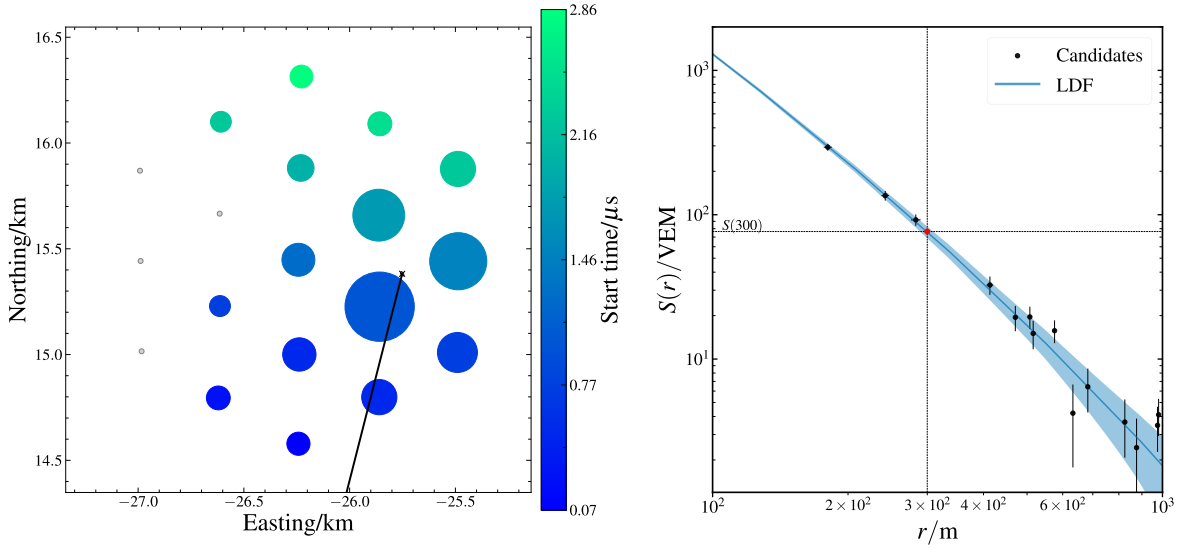


Figure 2.1: Example reconstruction of an SD-433 event with zenith angle $\theta = 32.0°$ and reconstructed shower size $S(300) = (76.5 \pm 2.9)$ VEM. The left panel shows the event footprint, with marker size proportional to the recorded signal and colour indicating the signal start time. The cross marks the reconstructed shower core, and the black line shows the projection of the arrival direction onto the ground. The right panel shows the recorded station signals as a function of the distance from the shower axis, together with the fitted LDF. The red point indicates the reconstructed shower-size estimator $S(300)$.

The slope of the LDF was determined on an event-by-event basis using a set of high-quality events with at least five triggered stations, for which β could be left as a free parameter in the fit [13]. The fitted values were then used to parametrize the average dependence of the slope on the shower size and zenith angle as

$$\beta(\theta, y) = a(\theta) + b(\theta) y \tag{2.6}$$

where $y = \lg[S(300)/\mathrm{VEM}]$. The coefficients that govern the dependence in zenith angle, $a(\theta)$ and $b(\theta)$ are modeled as second degree polynomials in $\sec\theta$,

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -3.72 & 1.30 & 0.055 \\ 0.98 & -1.30 & 0.385 \end{pmatrix} \begin{pmatrix} 1 \\ \sec\theta \\ \sec^2\theta \end{pmatrix}. \tag{2.7}$$

The parametrization describes the event-by-event fitted slopes with an average relative difference of approximately 2% over the considered range of shower sizes and zenith angles [13]. Once determined, this parametrization was fixed in the reconstruction of the complete data set, allowing events with insufficient station multiplicity to constrain the LDF slope to be reconstructed with the average LDF shape. With the LDF shape fixed by the parametrization of β, the shower size $S(300)$ and the core position $\vec{x}_c$ are determined simultaneously through a maximum-likelihood fit. The likelihood accounts for the measured signals in the triggered stations and for active stations that did not trigger. Although no signal trace is available for these non-triggered stations, their presence constrains the fit, since an LDF predicting a signal above the trigger threshold at their positions is disfavoured. In this way, the LDF fit determines $S(300)$ and refines the core position obtained from the initial geometrical reconstruction.

The left panel of Fig. 2.2 shows the distribution of the reconstructed shower size. At low values of $S(300)$, the number of events is strongly suppressed by the trigger efficiency.

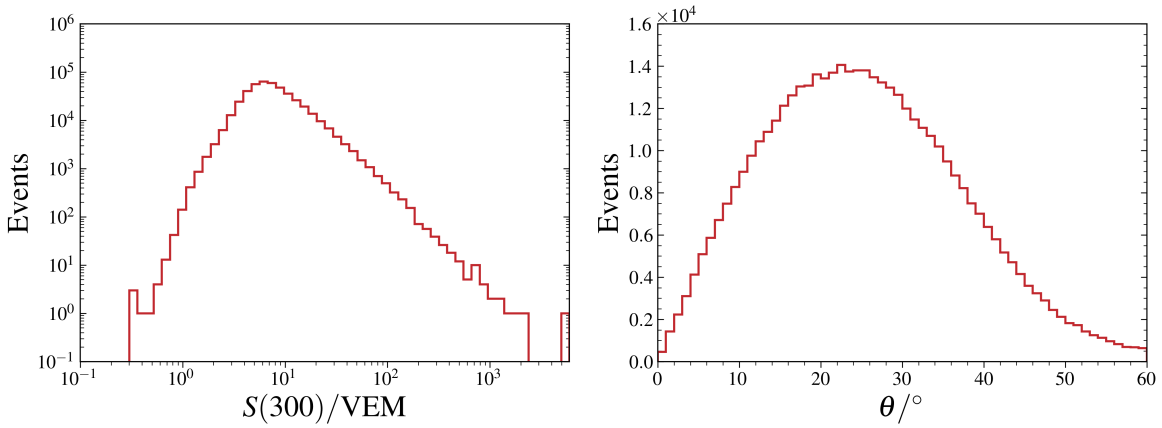


Figure 2.2: Distributions of the reconstructed shower size $S(300)$ (left) and zenith angle (right) for SD-433 events satisfying the 6T5 condition, with $0^\circ \leq \theta < 60^\circ$.

It increases as the array approaches full efficiency and then the distribution decreases approximately as a power law, reflecting the steeply falling cosmic-ray flux. The right panel shows the zenith angle distribution for the reconstructed events from 0 to 60°, the maximum angle of the vertical reconstruction implemented in Offline. The distributions contain 443 535 reconstructed events. However, as discussed in Section 1.2, events with $\theta < 45^\circ$ are used in the spectrum in order to lower the full efficiency threshold for the array.

In Fig. 2.3 the event multiplicity is shown in two $S(300)$ intervals, below and above 15 VEM. This threshold is chosen to roughly correspond to the signal of the full efficiency threshold. For the lower $S(300)$ interval, most events have a multiplicity of three or four. For the higher $S(300)$ interval, which roughly corresponds to the data set used for the spectrum, 98.3% of the events have a multiplicity greater than five. This improves the quality of the reconstructed events and provides a robust measurement of the signal and the energy of the events.

2.2 Correction for zenith-angle attenuation

The reconstructed signal $S(300)$ provides an estimator of the shower size at ground and contains information about the energy and longitudinal development of the primary cosmic-ray shower. However, at a fixed primary energy, $S(300)$ also depends on the zenith angle. Inclined showers traverse a larger atmospheric slant depth before reaching the detector and are therefore observed at a later stage of their development than more vertical showers of the same energy. Since the electromagnetic component attenuates more rapidly in the atmosphere than the more penetrating muonic component, their relative contributions to the signal at ground change with zenith angle. This effect is coupled to the different responses of the water-Cherenkov detector to electromagnetic particles and muons. Consequently, showers with the same primary energy but different zenith angles do not generally produce the same $S(300)$, and this dependence must be corrected before the shower size can be used as an energy estimator. The correction is determined from data using the constant-intensity-cut method, first introduced by Hersil et al. [15]. The corrected shower size, S_{30}, is the value of $S(300)$ that the shower would have produced at the reference angle $\theta_{\text{ref}} = 30^\circ$. This angle is chosen as the median of the studied event distribution, reducing the typical magnitude of

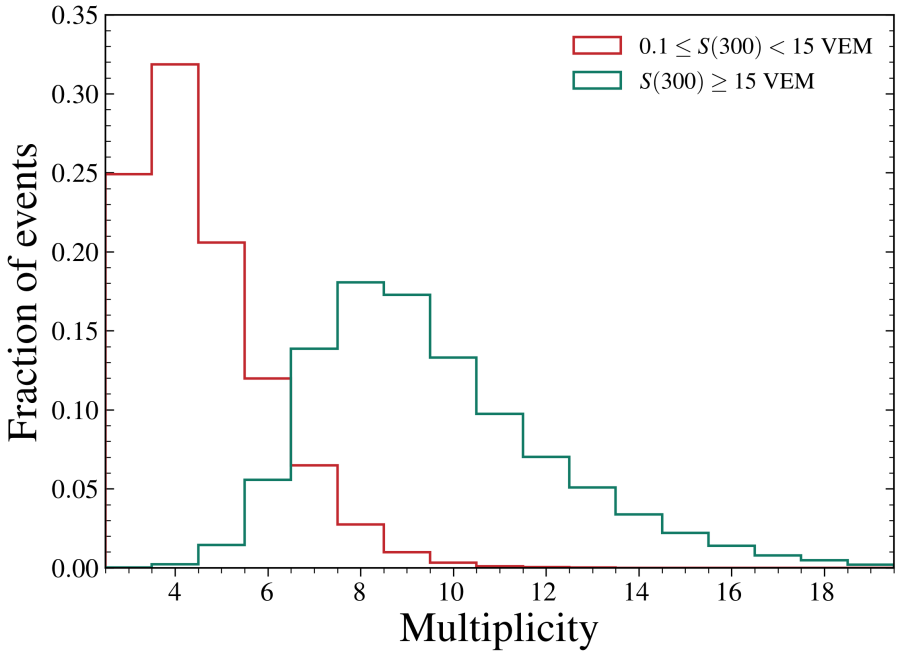


Figure 2.3: Multiplicity of reconstructed events below and above $S(300) = 15\,\text{VEM}$. Each distribution is normalized independently to show the change in its shape with shower size.

the correction. The measured and corrected shower sizes are related through the attenuation function,

$$S(300) = S_{30} f_{\text{CIC}}(\theta, S_{30}) \tag{2.8}$$

which is normalized such that $f_{\text{CIC}}(30^\circ, S_{30}) = 1$.

2.2.1 The Constant-intensity cut method

To determine the attenuation function, the Constant Intensity Cut (CIC) method is used. This method is based on the isotropy of arribal directions of the cosmic rays. The large-scale dipolar anisotropy reported by the Pierre Auger Observatory becomes significant above 8 EeV [16], while below this energy the flux can be considered isotropic. The events measured with the SD-433 are binned in equal-width intervals of $\sin^2\theta$, which have equal exposure. For this array, the zenith angle range $0^\circ \leq \theta < 45^\circ$ is divided into nine such intervals. Within each interval, events are ranked by decreasing $S(300)$. Events at the same rank N correspond approximately to the same primary energy, and the variation of their $S(300)$ with zenith angle determines the attenuation curve at that intensity. Six intensity cuts, $N \in \{100, 200, 500, 1000, 2000, 5000\}$, are used to determine the attenuation at different shower sizes. The statistical uncertainty of S_N is estimated from the Poisson fluctuations of N as

$$\sigma_{S_N} = \frac{S_{N-\sqrt{N}} - S_{N+\sqrt{N}}}{2}. \tag{2.9}$$

The attenuation also depends on the shower size, since the depth of the shower maximum and the contributions of the electromagnetic and muonic components evolve with energy. This dependence is included using S_{30} as an energy estimator. The fit uses $x = \sin^2\theta - \sin^2\theta_{\text{ref}}$ and $y = \lg(S_{30}/S_0)$, where the reference scale $S_0 = 40\,\text{VEM}$ is introduced to reduce

Table 2.1: Parameters obtained from the global fit of the SD-433 attenuation function.

Parameter	Fitted value
a_0	-1.6294 ± 0.0322
a_1	0.2623 ± 0.0589
b_0	-1.4622 ± 0.1328
b_1	-0.4461 ± 0.4125
c_0	0.8954 ± 0.7462

correlations among the parameters. The attenuation function is described by a third-degree polynomial in x, with coefficients that depend linearly on y,

$$f_{\mathrm{CIC}}(\theta, S_{30}) = 1 + (a_0 + a_1 y)x + (b_0 + b_1 y)x^2 + c_0 x^3. \tag{2.10}$$

First, each intensity is fitted independently. The resulting parameters are then used as starting values for a simultaneous fit to all six intensities and nine zenith-angle bins. The global fit gives $\chi^2/N_{\mathrm{dof}} = 31.80/43$. The best fit parameters are shown in Table 2.1.

Since f_{CIC} depends on the unknown S_{30}, the correction is applied iteratively, starting from $S_{30} = S(300)$ and updating it as $S_{30} = S(300)/f_{\mathrm{CIC}}(\theta, S_{30})$ until its relative change is smaller than 10^{-4}. The constant-intensity cuts and the global fit are shown in the upper panel of Fig. 2.4. The lower panel shows the corresponding attenuation curves, illustrating their dependence on S_{30}. Compared with the attenuation curve at the reference size $S_{30} = 40\,\mathrm{VEM}$, the shower-size dependence changes the reconstructed S_{30} by less than 3.5%, with the largest effect occurring for vertical showers.

The effect of the CIC correction is shown in Fig. 2.5. The upper panel shows the distribution of $S(300)$ as a function of the zenith angle. For a fixed intensity, the shower size decreases towards larger zenith angles due to atmospheric attenuation, following the fitted CIC curve. After the correction, the corresponding distributions in S_{30} become approximately independent of the zenith angle, as shown in the lower panel. The solid and dashed lines indicate the approximate full-efficiency threshold before and after the correction, respectively. Below this threshold, the distribution is distorted by the decreasing detection efficiency and should not be interpreted as a residual attenuation effect. The corrected shower size S_{30} is therefore used as the energy estimator.

2.3 Calibration of the SD-433 energy estimator

With the attenuation in the atmosphere corrected, the shower size S_{30} can be calibrated in terms of the primary energy. The other two surface-detector arrays, the SD-750 and SD-1500, are calibrated directly using events detected simultaneously by the SD and Fluorescence Detector (FD). However, because of the distance between the SD-433 and the nearest FD site, Cohihueco, the High Elevation Auger Telescopes (HEAT) telescopes cannot observe showers falling within the SD-433 area. The SD-433 energy scale is therefore obtained through an SD-SD calibration using events detected in coincidence with the SD-750. This introduces an additional complication because the SD-750 calibration relation must be extrapolated by approximately two to three energy bins below the range in which that array is fully efficient, as discussed in **??**. The finite resolutions of both arrays, their correlation, and the event selection must consequently be included in the calibration fit.

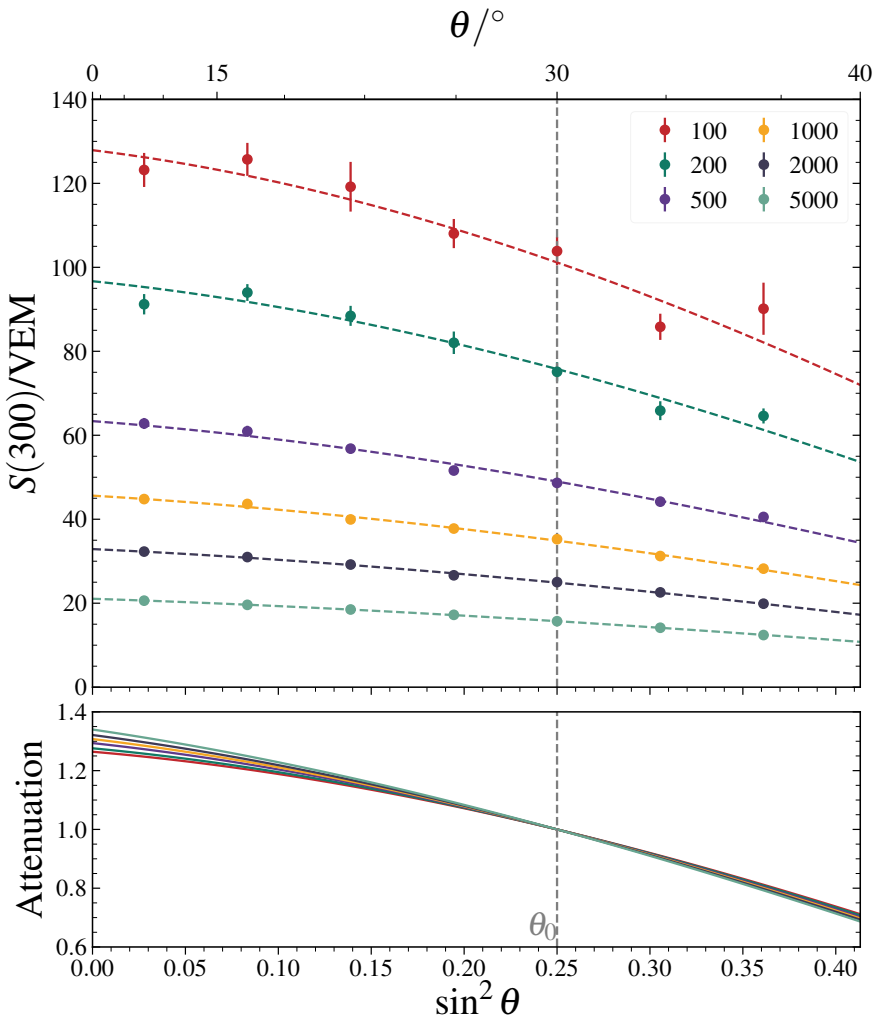


Figure 2.4: Global fit to the six constant-intensity cuts. The upper panel shows the values of $S(300)$ and the fitted curves, while the lower panel shows the corresponding attenuation functions. The vertical dashed line indicates the reference angle $\theta_{\text{ref}} = 30^\circ$.

2.3.1 SD-750 reference scale and coincident events

The calibration requires events reconstructed with high quality by both arrays. The standard spectrum-quality selections are therefore applied independently to the SD-433 and SD-750 datasets. The events are required to have $\theta < 40^\circ$, satisfy the 6T5 fiducial condition, and fall outside the periods of unstable acquisition. The SD-750 dataset is obtained from an updated Herald reconstruction of the data used in Ref. [8], restricted to the same acquisition period as the SD-433 dataset. Its energies are recalculated using the updated ICRC 2025 calibration based on the revised FD reconstruction [17, 18].

The SD-750 energy calibration is described by the power law

$$E^{\text{SD-750}} = A^{\text{SD-750}} \left(\frac{S_{35}^{\text{SD-750}}}{S_0^{\text{SD-750}}} \right)^{B^{\text{SD-750}}}, \tag{2.11}$$

where $S_{35}^{\text{SD-750}}$ is the corrected shower-size estimator, $A^{\text{SD-750}}$ sets the energy scale, and $B^{\text{SD-750}}$ is the slope. The values applied in this work are $A^{\text{SD-750}} = 13.2912\,\text{PeV}$, $B^{\text{SD-750}} = 0.9985676$,

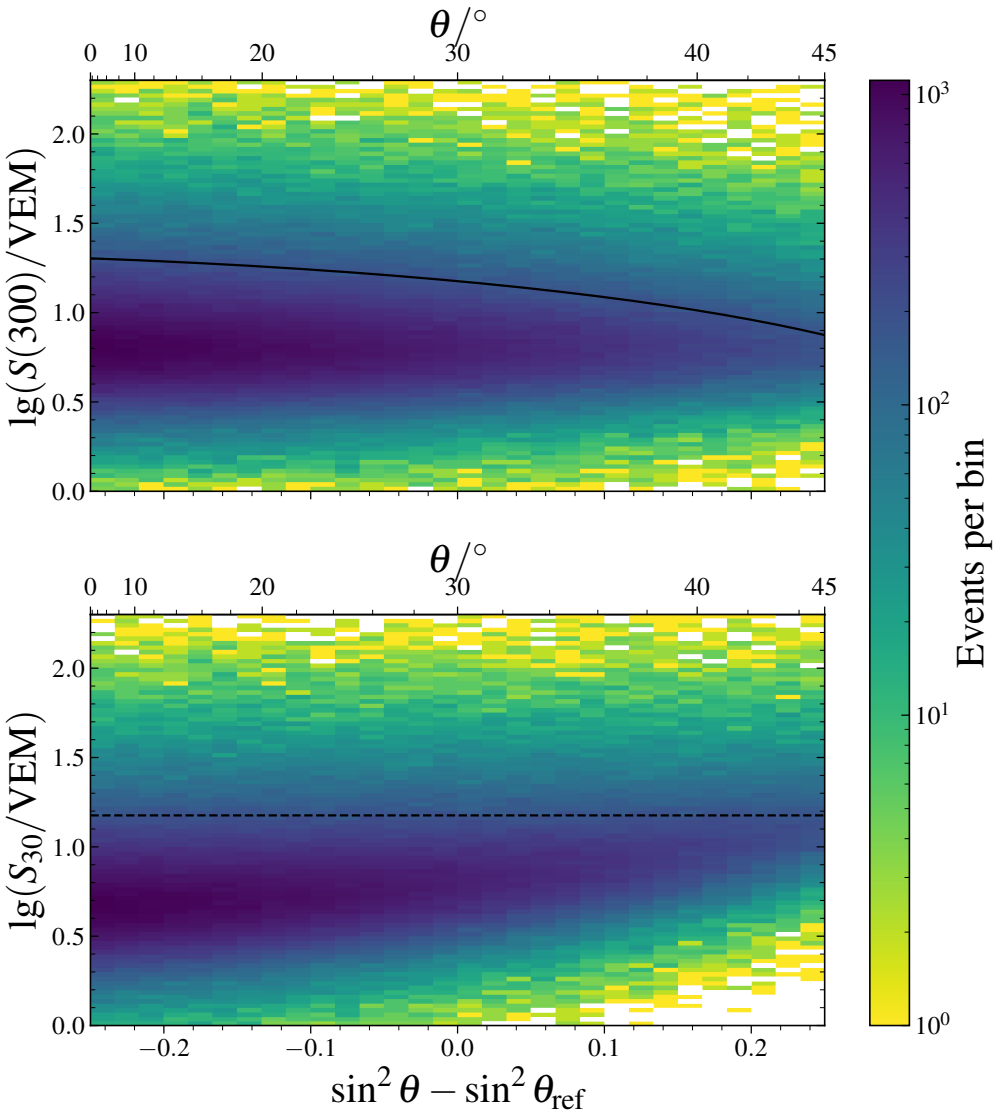


Figure 2.5: Distribution of the shower size and zenith angle before and after the CIC correction. The solid line in the upper panel shows the attenuation curve corresponding to the full-efficiency threshold, $S_{30} = 15\,\text{VEM}$. The dashed line in the lower panel shows the same threshold after the conversion to S_{30}.

and $S_0^{\text{SD-750}} = 1\,\text{VEM}$. After recalibrating the SD-750 events, the two datasets are matched by their event identifiers, yielding 37 505 coincident events.

2.3.2 Likelihood fit and calibration relation

The relation between the SD-433 shower size and the SD-750 reference energy is also described by a power law,

$$E = A \left(\frac{S_{30}}{S_0} \right)^B , \tag{2.12}$$

where $S_0 = 40\,\text{VEM}$ is chosen to be within the calibrated energy range to reduce the correlation between the fitted normalization A and slope B. The calibration is performed using the likelihood method introduced by Dembinski et al. [19]. The finite resolutions of both S_{30} and the SD-750 reference energy would bias a conventional χ^2-fit and are therefore included in

the likelihood. Unlike an SD-FD calibration, the two measurements are correlated because both arrays observe the same shower and share some of the water-Cherenkov detector (WCD)s. For the SD-750 resolution, the same parameterization from Ref. [8] is adopted,

$$\frac{\sigma_{\mathrm{SD-750}}}{E} = \sqrt{\sigma_{\mathrm{sh}}^2 + \left(\sigma_{\mathrm{det}}^{\mathrm{SD-750}}\right)^2 \frac{E_0}{E}}, \tag{2.13}$$

where $\sigma_{\mathrm{sh}} = 0.09$ describes the energy-independent shower-to-shower fluctuations, $\sigma_{\mathrm{det}}^{\mathrm{SD-750}} = 0.21$ parameterizes the detector contribution, and $E_0 = 100\,\mathrm{PeV}$.

Because both arrays sample the same shower, their shower-to-shower fluctuations are fully correlated. An additional detector correlation arises from the WCDs shared by the two arrays. The S_{30} resolution is estimated through the inverse calibration using an initial slope $B_0 = 1$,

$$\frac{\sigma_{S_{30}}}{S_{30}} = \frac{1}{B_0}\sqrt{\sigma_{\mathrm{sh}}^2 + \left(\sigma_{\mathrm{det}}^{\mathrm{SD-433}}\right)^2 \frac{E_0}{E}}, \tag{2.14}$$

where $\sigma_{\mathrm{det}}^{\mathrm{SD-433}} = 0.12$. The combined response function is then modelled as a correlated two-dimensional Gaussian. An underlying shower size s is therefore introduced, allowing the reconstructed S_{30} to fluctuate around it according to the SD-433 resolution model. For a given s, the calibration relation predicts the energy $E_{\mathrm{cal}}(s \mid A, B) = A(s/S_0)^B$. The same reasoning is also applied to the SD-750 energy, allowing the reconstructed $E_i^{\mathrm{SD-750}}$ to fluctuate around $E_{\mathrm{cal}}(s \mid A, B)$ according to the SD-750 resolution model. The measured pair $(S_{30,i}, E_i^{\mathrm{SD-750}})$ is therefore described by a correlated two-dimensional Gaussian.

The corresponding response function for event i is

$$R_i\left(S_{30,i}, E_i^{\mathrm{SD-750}} \mid s, A, B\right) = \frac{\exp\left[-Q_i(s \mid A, B)/2\right]}{2\pi\sigma_{S,i}\sigma_{E,i}\sqrt{1-\rho_i^2}}, \tag{2.15}$$

where $\sigma_{S,i}$ and $\sigma_{E,i}$ are the absolute uncertainties of $S_{30,i}$ and $E_i^{\mathrm{SD-750}}$, respectively. The energy-dependent correlation coefficient is calculated as

$$\rho_i = \frac{\dfrac{\sigma_{\mathrm{sh}}^2}{B_0} + \rho_{\mathrm{det}}\sigma_{\mathrm{det}}^{\mathrm{SD-433}}\sigma_{\mathrm{det}}^{\mathrm{SD-750}}\dfrac{E_0}{E_i^{\mathrm{SD-750}}}}{\left(\dfrac{\sigma_{S,i}}{S_{30,i}}\right)\left(\dfrac{\sigma_{E,i}}{E_i^{\mathrm{SD-750}}}\right)}, \tag{2.16}$$

where $\rho_{\mathrm{det}} = 0.33$ is the assumed correlation between the detector contributions arising from the WCDs shared by the two arrays. The quantity Q_i is the quadratic correlated distance between the measured pair and the corresponding point on the calibration curve,

$$Q_i(s \mid A, B) = \frac{1}{1-\rho_i^2}\left[\left(\frac{S_{30,i}-s}{\sigma_{S,i}}\right)^2 + \left(\frac{E_i^{\mathrm{SD-750}} - E_{\mathrm{cal}}(s \mid A, B)}{\sigma_{E,i}}\right)^2 - 2\rho_i\left(\frac{S_{30,i}-s}{\sigma_{S,i}}\right)\left(\frac{E_i^{\mathrm{SD-750}} - E_{\mathrm{cal}}(s \mid A, B)}{\sigma_{E,i}}\right)\right]. \tag{2.17}$$

Because the event distribution falls steeply with shower size, upward migrations are more frequent than downward migrations. The distribution of the shower size is therefore included in the likelihood. An empirical distribution is used [20], defined as

$$h(s) = \frac{1}{N_{\mathrm{evt}}}\sum_{j=1}^{N_{\mathrm{evt}}}\delta(s - S_{30,j}) \tag{2.18}$$

where the sum is done over the observed events, N_{evt} is the total number of events and δ is Dirac's delta function. When binning in $\lg(S_{30}/\,\mathrm{VEM})$ this expression can be simplified to $w_k = n_k/N_{\mathrm{evt}}$, where n_k is the number of events in bin k. With this, the single event likelihood can be defined

$$\mathcal{L}_i(A,B) = \sum_k w_k R_i\left(S_{30,i}, E_i^{\mathrm{SD-750}} \mid s_k, A, B\right), \tag{2.19}$$

where the sum over k replaces the integral due to the binning in $\lg(s_k/\mathrm{VEM})$. Finally, the combined log-likelihood is written as $\ln\mathcal{L}(A,B) = \sum_i \ln\mathcal{L}_i(A,B)$. Since the SD-433 has a better resolution, the event selection is applied to S_{30} to minimize migration effects. The cut $S_{30} > 30\,\mathrm{VEM}$ roughly corresponds to the full-efficiency threshold of the SD-750, leaving 5980 available events for the calibration. However, events below the cut are still used to determine $h(s)$ and account for migrations across the threshold. The fit results in $A = (149.49 \pm 0.34)\,\mathrm{PeV}$ and $B = 0.963 \pm 0.002$, with a correlation $\rho_{AB} = -0.51$. The calibration threshold $S_{30} = 30\,\mathrm{VEM}$ corresponds to an energy of 113.3 PeV. The systematic uncertainty associated with the choice of the calibration threshold is evaluated by repeating the fit for $S_{30,\mathrm{min}}$ between 15 and 40 VEM. As the threshold increases, the normalization A decreases while the slope B increases, resulting in strongly anticorrelated variations. The full excursion of the fitted parameters is assigned as the systematic uncertainty, yielding $\Delta A_{\mathrm{syst}} = 4.0\,\mathrm{PeV}$ and $\Delta B_{\mathrm{syst}} = 0.035$. The fit with $S_{30,\mathrm{min}} = 15\,\mathrm{VEM}$ extends well below the nominal SD-750 efficiency range and approximately down to the full-efficiency threshold of the SD-433, therefore providing a conservative estimate of the systematic envelope.

The relative difference between the SD-750 reference energy and the calibrated SD-433 energy is shown in the left panel of Fig. 2.7. The small points correspond to individual events. Events below the $S_{30,\mathrm{min}} = 30\,\mathrm{VEM}$ shower-size threshold are shown in gray, while the events used for the SD-433 calibration are shown in green. The red points show the mean residual in bins of width $\Delta\lg(E^{\mathrm{SD-433}}/\mathrm{eV}) = 0.1$ below $\lg(E^{\mathrm{SD-433}}/\mathrm{eV}) = 18$ and 0.3 above to increase the statistics. The error bars indicate the standard error of the mean. Although individual residuals can reach approximately 20%, above the calibration threshold, the mean residual remains close to zero and decreases quickly with energy. The positive residual below the threshold is produced by migration across the event selection and is not included in the fit.

Finally, the energy distribution of the SD-433 events is shown in the right panel of Fig. 2.7, together with the full-efficiency threshold. Below this threshold, the detection efficiency increases with energy, producing the rising event distribution despite the decreasing cosmic ray flux. Above full efficiency, the distribution decreases approximately linearly on the logarithmic scale, as expected for a power-law-like spectrum. This distribution forms the basis for the raw spectrum determined in Section 3.1 and the subsequent characterization of the second knee.

2.4 Energy resolution and bias

The energy resolution and bias enter the detector response used to fit and unfold the cosmic-ray spectrum measured with the SD-433. Their parameterizations are obtained from the same simulations used to determine the detection efficiency in Section 1.2, adopting the mixed composition inferred from the GSF model [21]. The denser spacing of the SD-433 provides more triggered stations at a given energy, better constraining the shower-size reconstruction and yielding a better energy resolution than that of the SD-750. Comparisons using events reconstructed jointly by the two arrays provide a consistency check above full

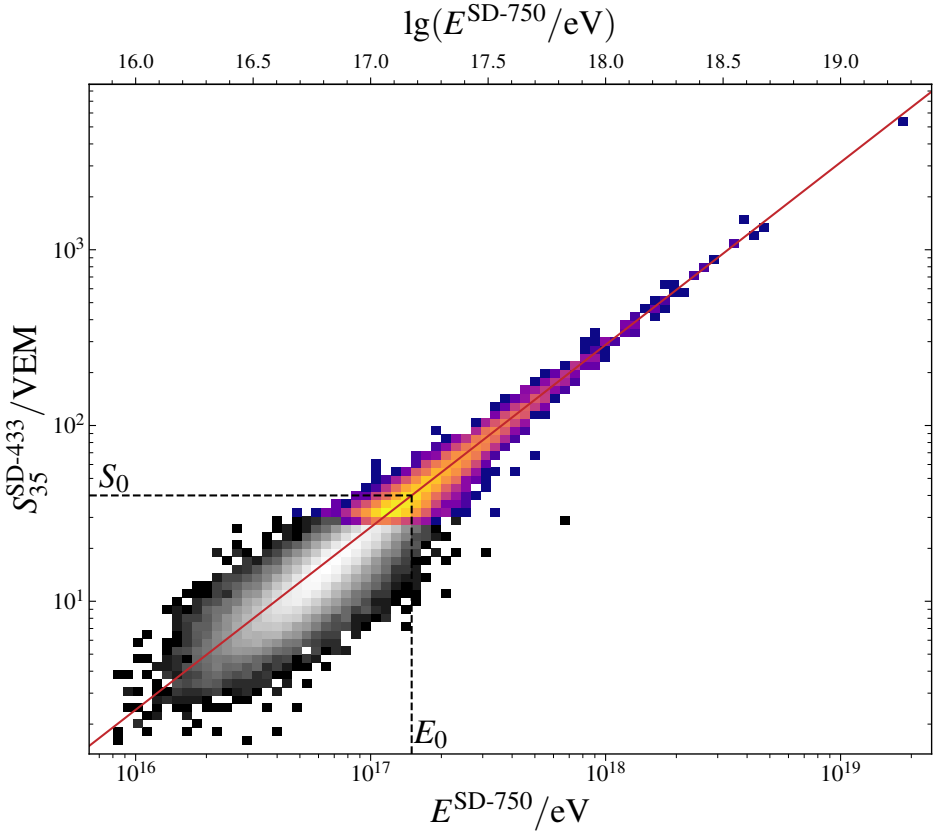


Figure 2.6: Calibration of the SD-433 shower size S_{30} against the SD-750 reference energy using coincident events. The color scale shows the event density above the calibration threshold, while events below the threshold are shown in gray. The red line shows the fitted calibration relation. The dashed lines indicate the normalization point $S_0 = 40\,\mathrm{VEM}$ and its corresponding energy.

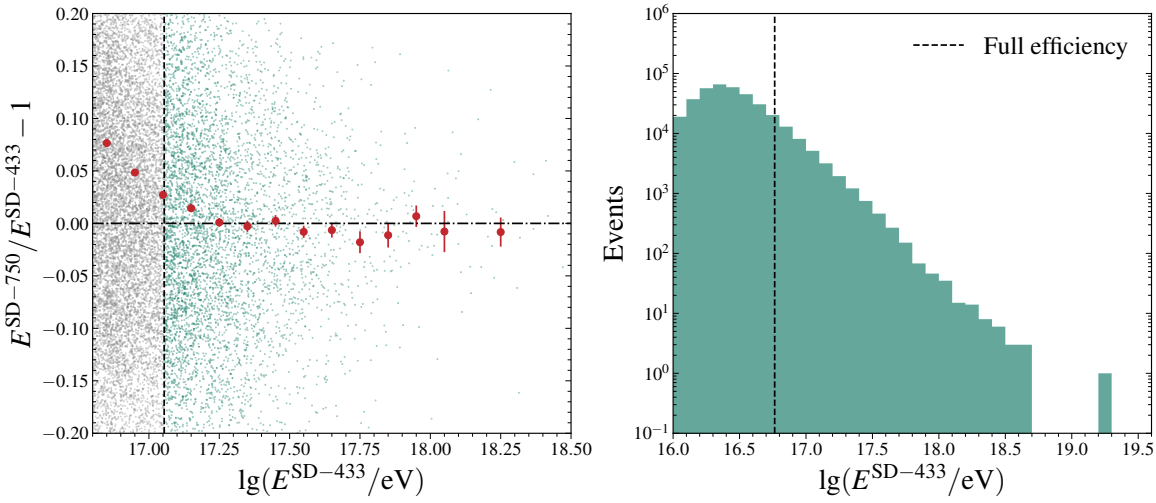


Figure 2.7: Left: relative difference between the SD-750 reference energy and the calibrated SD-433 energy. The points show individual events and the red markers show the mean residual in energy bins, with error bars indicating the uncertainty of the mean. Right: energy distribution of the full SD-433 dataset. The dashed line indicates the full-efficiency energy.

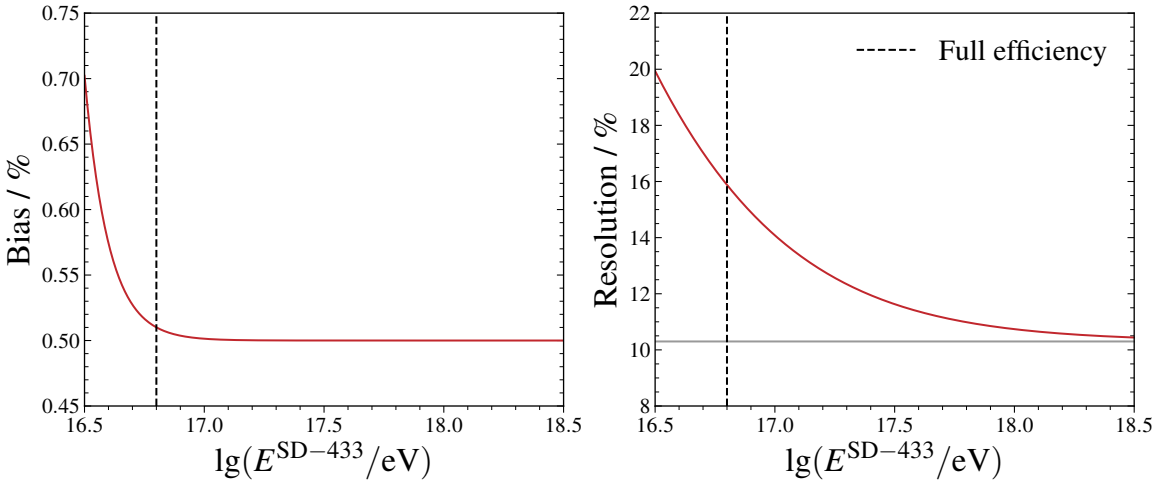


Figure 2.8: Energy bias (left) and resolution (right) parameterizations obtained from simulations using the GSF mixed composition. The dashed lines indicate the full-efficiency threshold. Adapted from Ref. [21].

efficiency, although the direct determination is limited by the available statistics and the unknown correlation between the energy estimators.

The relative energy bias is determined from the mean fractional residual $E_{\mathrm{rec}}/E_{\mathrm{MC}} - 1$. These residuals are evaluated using the simulation-based calibration, while the energy dependence is expressed in terms of the true energy mapped onto the SD-433 data scale, denoted here by E. The bias is parameterized as

$$b(E) = b_0 \exp[-b_1 \,(\lg(E/\mathrm{eV}) - 16)] + b_2, \tag{2.20}$$

with

$$b_0 = 0.30 \pm 0.01, \qquad b_1 = 10.0 \pm 0.3, \qquad b_2 = 0.005 \pm 0.001. \tag{2.21}$$

Below full efficiency, upward fluctuations are preferentially selected, producing a positive bias that increases towards lower energies. Above the full-efficiency threshold, this contribution becomes small.

The relative energy resolution is described by the quadratic sum of a constant contribution from shower-to-shower fluctuations and an energy-dependent detector contribution,

$$\frac{\sigma(E)}{E} = \sqrt{R_{\mathrm{sh}}^2 + \frac{R_0^2}{E/(10^{17}\,\mathrm{eV})}}, \tag{2.22}$$

where the fit to the simulated GSF mixture yields

$$R_{\mathrm{sh}} = 0.103 \pm 0.001, \qquad R_0 = 0.096 \pm 0.005. \tag{2.23}$$

The detector contribution decreases with increasing energy, and the relative resolution approaches the shower-fluctuation limit of approximately 10.3%. Here, $\sigma(E)$ denotes the absolute energy resolution used in the response function for the spectrum analysis. The bias and relative resolution parameterizations are shown in Fig. 2.8.

2.5 Summary

In this chapter we present the energy reconstruction of the events recorded with the SD-433. After reconstructing the shower geometry, $S(300)$ is obtained by modelling the

decrease of the signal with the distance to the shower core. From the reconstructed $S(300)$, the attenuation function is determined to correct for the different zenith angles of the showers. The zenith-independent shower size S_{30} is then obtained by iteratively applying the shower-size-dependent correction $f_{\mathrm{CIC}}(\theta, S_{30})$. Using the events detected in common with the SD-750 and the zenith-independent shower size estimator we then developed the energy calibration for the SD-433. The likelihood formalism was introduced to take into account the finite resolution both of SD-433 in the shower size and SD-750 in energy. An empirical shower-size distribution is included to account for the preferential migration from low to high energies produced by the steep cosmic ray spectrum. Minimizing the log-likelihood yields $A = (149.49 \pm 0.34)$ PeV and $B = 0.963 \pm 0.002$. The mean difference between the calibrated SD-433 energy and the SD-750 reference energy remains below 3% throughout the fitted energy range and decreases with increasing energy. Finally, the calibrated SD-433 energy distribution forms the basis of the cosmic ray spectrum measurement presented in the next chapter.

Chapter 3

The SD-433 Energy spectrum

With the developed energy reconstruction from the previous chapter and the determination of the full efficiency to have a proper estimation of the exposure, this chapter focuses on the measurement of the cosmic ray spectrum with the SD-433. For this, Section 3.1 is devoted to the construction of the raw spectrum. After this, the forward-folding formalism is presented in order to fit the spectrum and measure the second knee in the cosmic ray spectrum. Several checks were developed to assess the robustness of the method and to understand the impact of the different detector characteristics. In Section 3.3, the unfolded spectrum is shown and the robustness of the measurement of the second knee is studied. Finally, in Section 3.4 the impact of the different sources of systematics is assessed to give a complete picture of the measurement of the spectrum and the second knee.

3.1 Raw spectrum

The energy spectrum describes the differential flux of cosmic rays as a function of their energy. It is estimated from the number of events recorded in a given energy interval, normalized by the exposure of the detector and the width of the interval. For the SD-433, the dataset described in Chapter 1 contains 34 406events that pass the 6T5 fiducial trigger with zenith angle lower than 40°. These events were recorded between 1 July 2018 and 21 December 2021 above the full-efficiency threshold.

The raw flux in the bin i of reconstructed-energy is estimated as

$$J_i^{\text{raw}} = \frac{N_i}{\mathcal{E}\,\Delta E_i}, \tag{3.1}$$

where N_i is the number of events in the bin, ΔE_i is the bin width, and $\mathcal{E}$ is the accumulated exposure. As determined in Section 1.4, the exposure is $2.85\,\text{km}^2\,\text{sr}\,\text{yr}\pm0.09$ and is energy independent above the full-efficiency threshold. A logarithmic bin width of $\Delta\lg(E/\text{eV}) = 0.1$ is used up to $\lg(E/\text{eV}) = 18.1$, chosen to be compatible with the energy resolution described in Section 2.4. Above this energy, a coarser binning of $\Delta\lg(E/\text{eV}) = 0.3$ is adopted to increase the number of events per bin and, therefore, reduce the statistical uncertainties. This spectrum is referred to as the *raw spectrum* because no correction has yet been applied for event migrations between energy bins caused by the finite resolution and bias of the detector.

The left panel of Fig. 3.1 shows the raw flux together with the 68% Poisson confidence intervals, which are only visible in the bins with fewer events. The number of events in each bin is indicated next to each data point. Since the cosmic-ray flux decreases approximately as E^{-3}, the same measurements are shown multiplied by E^3 in the right panel. This representation reduces the dominant power-law dependence and makes changes in the spectral

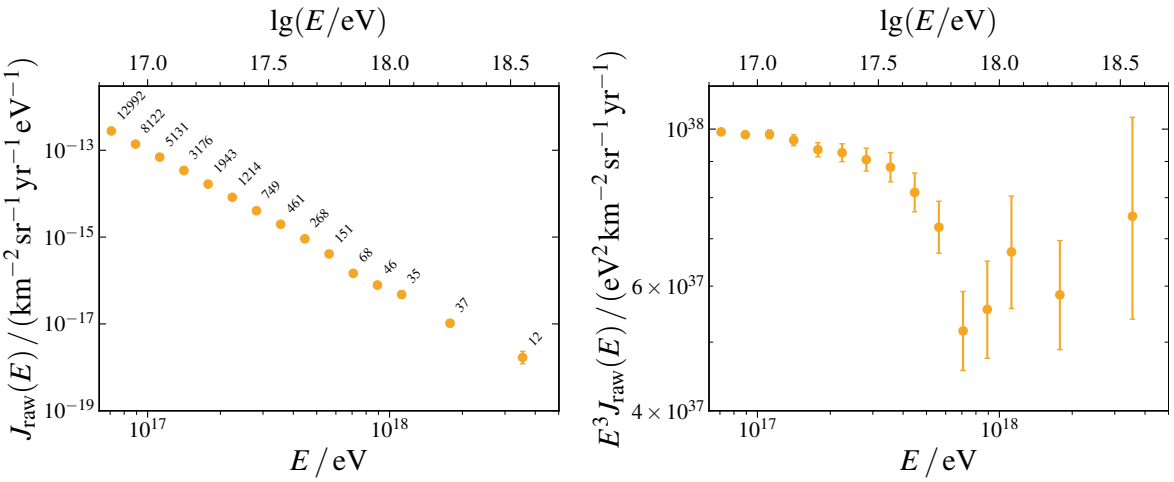


Figure 3.1: Raw cosmic-ray spectrum measured with the SD-433. The error bars represent the 68% Poisson confidence intervals. Left: differential flux as a function of reconstructed energy; the numbers indicate the number of events in each bin. Right: the same flux multiplied by E^3 to highlight changes in spectral slope.

slope more visible. A steepening around $\lg(E/\mathrm{eV}) \simeq 17.3$ already suggests the presence of a feature compatible with the second knee, although the detector response must be taken into account before it can be properly measured.

3.2 Forward folding

The raw spectrum is affected by the finite energy resolution, the energy bias, and the detection efficiency of the SD-433. These effects must be included when fitting the spectrum to determine the spectral parameters because they cause events to migrate between reconstructed-energy bins. A direct inversion of the detector response can amplify statistical fluctuations [22, 23]. Some experiments therefore use unfolding procedures that require external regularization to control these effects. Instead, a forward-folding procedure is used here. For a spectrum model $J(E)$, the expected event distribution is folded through a convolution with the detector response and compared directly with the measured counts.

The detector response is represented by a migration matrix,

$$R_{ij} = \frac{\displaystyle\int_{\Delta E'_i} \mathrm{d}E' \int_{\Delta E_j} \mathrm{d}E\, J(E)\, \epsilon(E)\, \kappa(E' \mid E)}{\displaystyle\int_{\Delta E_j} \mathrm{d}E\, J(E)}, \tag{3.2}$$

where E is the true energy and E' the reconstructed energy. The detector response $\kappa(E_{\rm rec} \mid E)$ represents the conditional probability density for reconstructing an energy $E_{\rm rec}$ given a true energy E. The detector efficiency $\epsilon(E)$, determined in Section 1.2, weighs the probability that an event is detected. Above full efficiency, elements R_{ij} represent the probability that an event with true energy in bin j is reconstructed in bin i. However, at lower energies, where the efficiency is smaller than unity, the matrix also accounts for the probability that an event is detected, allowing the contribution from true energies below the full-efficiency threshold to be included.

The response function is modeled as a Gaussian:

$$\kappa(E_{\mathrm{rec}} \mid E) = \frac{1}{\sqrt{2\pi}\,\sigma(E)} \exp\left[-\frac{(E_{\mathrm{rec}} - E\,[1 + b(E)])^2}{2\sigma^2(E)}\right], \tag{3.3}$$

where $\sigma(E)$ is the energy resolution and $b(E)$ is the relative energy bias determined from the simulations in Section 2.4.

At high energies, where the relative resolution $\sigma(E)/E$ is dominated by shower-to-shower fluctuations, the distribution of events among neighbouring reconstructed-energy bins becomes approximately independent of energy. Consequently, the width and shape of the migration pattern in logarithmic energy are approximately constant. For a spectrum model with parameters $\mathbf{s}$, the expected number of events in the true-energy bin j is

$$\mu_j(\mathbf{s}) = \int_{\Delta E_j^{\mathrm{true}}} \mathrm{d}E\, J(E \mid \mathbf{s})\, \mathcal{E}(E), \tag{3.4}$$

where $\mathcal{E}(E)$ is the exposure. The spectrum is only calculated above full-efficiency to provide a robust estimation of the number of events becomes independently of any efficiency model. After folding with the migration matrix, the expected number of events in the reconstructed-energy bin i is

$$\nu_i(\mathbf{s}) = \sum_j R_{ij}\, \mu_j(\mathbf{s}). \tag{3.5}$$

The spectrum parameters are determined by comparing the expected event number ν_i with the observed event number n_i in each reconstructed-energy bin using a binned Poisson likelihood. For this purpose, the Poisson deviance,

$$D(\mathbf{s}) = 2\sum_i \left[\nu_i(\mathbf{s}) - n_i + n_i \ln\left(\frac{n_i}{\nu_i(\mathbf{s})}\right)\right], \tag{3.6}$$

is minimized with respect to $\mathbf{s}$. It should be noted that the migration matrix in **??** explicitly depends on the spectrum model $J(E \mid \mathbf{s})$. Consequently, every change in the spectrum parameters $\mathbf{s}$ made during each fit iteration requires the matrix to be recalculated. To avoid this expensive calculation, an approximate migration matrix is used following the approach described in Appendix C of [24].

If the true-energy bins are sufficiently narrow, the efficiency and response function can be approximated as constant within each bin and taken outside the true-energy integral. The remaining integral of $J(E \mid \mathbf{s})$ in the numerator becomes identical to that in the denominator of **??**, and the dependence on $J(E \mid \mathbf{s})$ cancels out. This gives the approximate migration matrix

$$R_{ij}^{\mathrm{approx}} = \epsilon(E_j) \int_{\Delta E_i^{\mathrm{rec}}} \mathrm{d}E_{\mathrm{rec}}\, \kappa(E_{\mathrm{rec}} \mid E_j). \tag{3.7}$$

For the numerical calculation, the true-energy bin width is chosen as $\Delta \lg(E_j/\mathrm{eV}) = 0.01$, while the reconstructed-energy bin width is $\Delta \lg(E_i/\mathrm{eV}) = 0.1$. The approximate migration matrix for the SD-433 is shown in Fig. 3.2, together with the full-efficiency threshold in red and the identity in white. The horizontal axis represents the true energy, while the vertical axis represents the reconstructed energy. To avoid border effects, the migration matrix is calculated from 10^{16} to 10^{19} eV, extending below the full-efficiency threshold and above the maximum reconstructed energy in the dataset.

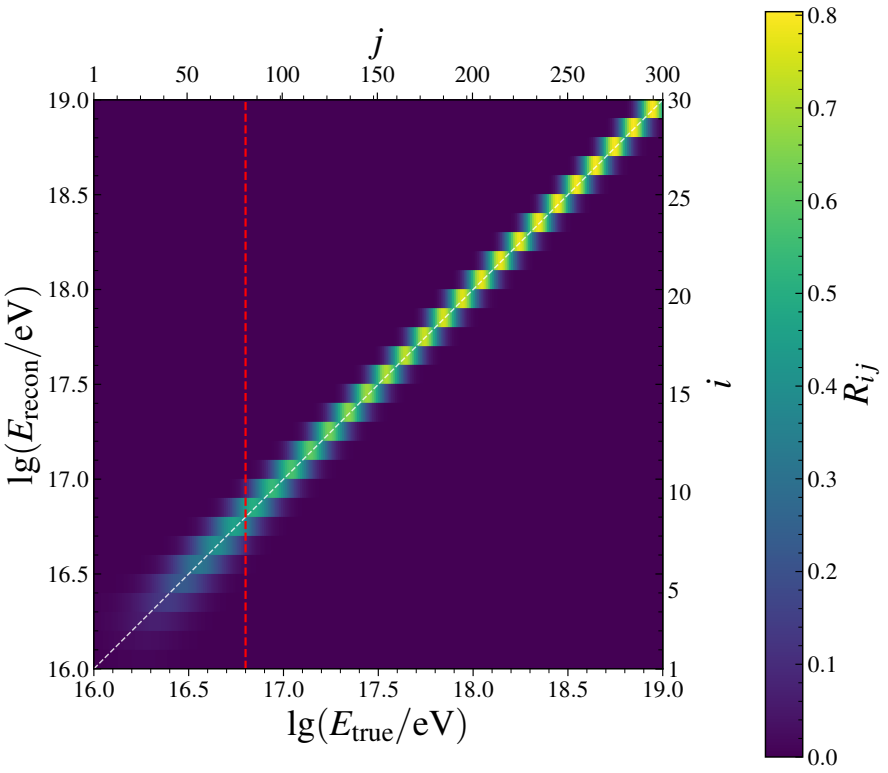


Figure 3.2: Approximate migration matrix for the SD-433. The horizontal axis represents true energy and the vertical axis reconstructed energy. The red dashed line indicates the full-efficiency threshold in true energy, and the white dashed line indicates $E_{\text{rec}} = E_{\text{true}}$.

Although the spectrum is only calculated for a reconstructed energy greater than the full efficiency threshold energy, events with a true energy below the threshold indirectly contribute to the spectrum prediction through upward energy fluctuations accounted for in the migration matrix. Given the energy resolution and the chosen bin width, the first sub-threshold bin and, to a lesser extent, the second one have some impact on the spectrum. To account for them, the efficiency shape before the full efficiency threshold is taken into consideration via the calculation of the number of events μ_j as per eq. 3.5.

The efficiency suppresses the contribution from energies below the full-efficiency threshold. Nevertheless, upward migrations allow events from up to two bins below the full-efficiency threshold to contribute significantly to the measured spectrum. At low energies, the worsening energy resolution broadens the migration pattern, increasing the contributions from more distant true-energy bins. At high energies, where shower-to-shower fluctuations dominate the relative energy resolution, the shape of the migration pattern becomes approximately independent of energy.

The next step is to evaluate the usefulness of this approximate matrix by comparing the expected event numbers in bins of reconstructed energy predicted with the exact and approximate matrices given the same number of events of true energy. To assess the accuracy of the approximation, the power-law spectrum,

$$J(E) = J_0 E^{\gamma}, \tag{3.8}$$

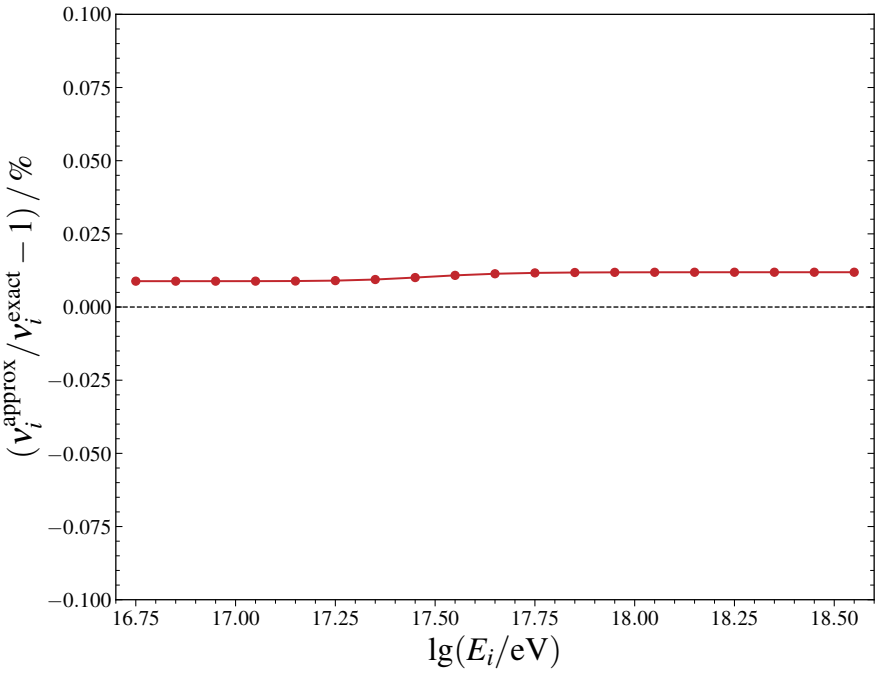


Figure 3.3: Relative difference between the expected event numbers obtained using the approximate and exact migration matrices for a power-law spectrum with $\gamma = -3$.

with spectral index $\gamma = -3$ is used. The expected event numbers ν_i^{approx} and ν_i^{exact} are calculated using the exact matrix, R_{ij}^{exact}, and the approximate matrix, R_{ij}^{approx}, respectively. The relative difference,

$$\delta\nu_i = \frac{\nu_i^{\mathrm{approx}} - \nu_i^{\mathrm{exact}}}{\nu_i^{\mathrm{exact}}}, \tag{3.9}$$

is shown in Fig. 3.3.

The difference remains below 0.02% over the full energy range, showing that the effect of the approximation is negligible for the chosen binning. The approximate matrix is therefore used in the forward-folding fit, avoiding both an a priori dependence the calculation of the migration matrix at every iteration and the postulation of a priori spectrum close enough to the final one to ensure numerical convergence.

The best-fit spectrum parameters $\hat{\mathbf{s}}$ are determined using the described procedure. Subsequently, the ratio,

$$c_i = \frac{\mu_i(\hat{\mathbf{s}})}{\nu_i(\hat{\mathbf{s}})}, \tag{3.10}$$

is calculated for each bin.

The corrections are less than 7% over the full energy range of the SD-433 spectrum as shown in Fig. 3.4. These correction factors are more relevant near the full-efficiency threshold, where the energy resolution has the greatest impact on the migration of events. At high energies, they become approximately constant because the relative shower-to-shower fluctuations, which are assumed to be independent of energy, dominate the energy resolution.

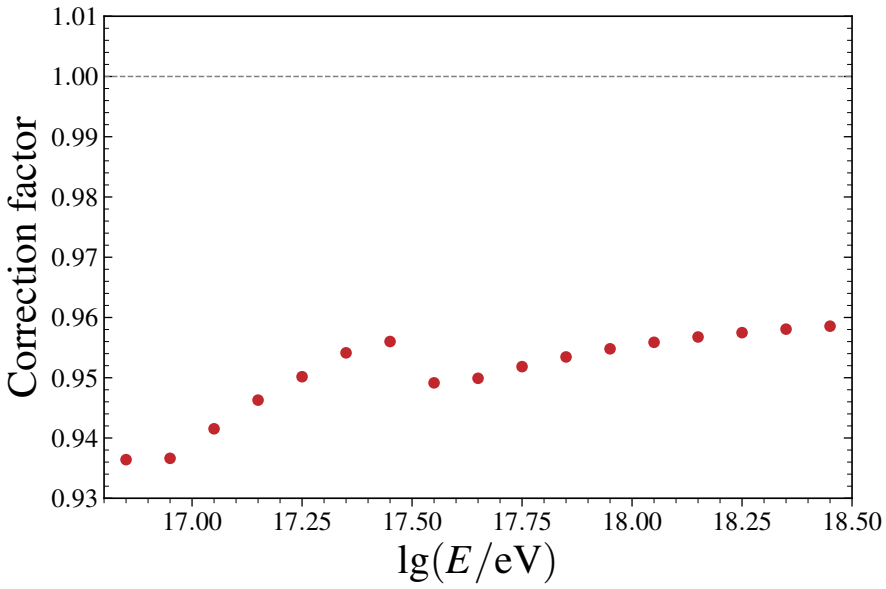


Figure 3.4: Correction factors obtained from the forward-folding fit to the measured SD-433 dataset.

3.3 The SD-433 Unfolded spectrum

The correction factors are then used to estimate the unfolded flux J_i from the measured one,

$$J_i = c_i J_i^{\text{raw}}. \tag{3.11}$$

The SD-433 spectrum is fitted between $\lg(E/\text{eV}) = 16.8$ and 18.5. The lower limit is determined by the full-efficiency threshold, and the upper limit is chosen below the ankle so that the second knee can be determined without requiring an additional spectral feature.

To test the presence of a knee-like structure in the spectrum, the data are first fitted with a simple power law,

$$J(E) = J_0 \left(\frac{E}{10^{17}\,\text{eV}} \right)^{-\gamma}, \tag{3.12}$$

where the normalization J_0 and spectral index γ are free parameters. The resulting fit, shown in **??**, yields a Poisson deviance of $D/\text{dof} = 41.1/15$, corresponding to a p-value of 3.1×10^{-4}. The simple-power-law hypothesis is therefore rejected, providing strong evidence for a change in the spectral slope within the measured energy range. As a consequence, a broken power law with a soft transition is used to fit the spectrum,

$$J(E) = J_0 \left(\frac{E}{10^{17}\,\text{eV}} \right)^{-\gamma_0} \left[1 + \left(\frac{E}{E_{01}} \right)^{\frac{1}{\omega_{01}}} \right]^{(\gamma_0-\gamma_1)\omega_{01}}, \tag{3.13}$$

where J_0 is a normalization factor. The second knee is characterized by its energy E_{01}, the spectral indices γ_0 and γ_1 before and after the feature, respectively, and ω_{01}, which governs the width of the transition [8].

The spectrum is first fitted with all five parameters left free. The fit converges and yields a Poisson deviance of $D/\text{dof} = 12.3/12$, corresponding to a p-value of 0.42. The fitted parameters are listed in Table 3.1. Although the second-knee energy and spectral indices are

Table 3.1: Spectral parameters and fit quality for the SD-433 spectrum obtained with ω_{01} free and fixed to 0.05 together with statistical uncertainties.

Parameter	Free ω_{01}	Fixed $\omega_{01} = 0.05$
$J_0 / \left(10^{-13}\,\mathrm{km^{-2}\,yr^{-1}\,sr^{-1}\,eV^{-1}}\right)$	0.913 ± 0.007	0.913 ± 0.005
$\lg(E_{01}/\mathrm{eV})$	17.44 ± 0.09	17.44 ± 0.08
γ_0	3.03 ± 0.03	3.04 ± 0.02
γ_1	3.34 ± 0.08	3.34 ± 0.08
ω_{01}	0.09 ± 0.27	0.05 (fixed)
D/dof	12.3/12	12.3/13
p-value	0.42	0.50

well constrained, the transition width is reconstructed as $\omega_{01} = 0.09 \pm 0.27$. The uncertainty is considerably larger than the fitted value, and the corresponding symmetric interval includes unphysical negative values, indicating that the SD-433 data cannot constrain the width of the transition.

Therefore, we perform a second fit fixing $\omega_{01} = 0.05$, the same value adopted by the Pierre Auger Observatory for spectral features whose transition widths could not be constrained [24]. This fit also provides a good description of the data. Leaving the transition width free keeps the deviance stable, despite introducing an additional fit parameter. The reconstructed second-knee energy and spectral indices are compatible between the two fits within their uncertainties. Therefore, the inability to constrain ω_{01} does not significantly affect the determination of the other spectral parameters. Consequently, the transition width is fixed for the spectrum measurement using the SD-433 alone. Although subdominant, the influence of ω_{01} on the measured spectrum is examined further in Section 3.4.4, together with the other sources of systematic uncertainty.

The resulting unfolded spectrum and the residuals with respect to the fitted model are shown in Fig. 3.5.

To assess the stability of the spectral parameters, the fit was repeated by progressively increasing the minimum fit energy from $\lg(E_{\mathrm{min}}/\mathrm{eV}) = 16.8$ to 17.2. The results are shown in Fig. 3.6. The reconstructed second-knee energy and spectral indices remain compatible within statistical uncertainties over the complete range. As the minimum energy increases, the number of events and the lever arm below the second knee decrease. Consequently, the uncertainty in γ_0 increases, although the fitted value remains compatible with the nominal result at $\lg(E_{\mathrm{min}}/\mathrm{eV}) = 16.8$.

3.4 Spectrum systematics

The systematic uncertainties affecting the SD-433 spectrum arise from the exposure, the energy scale, the detector response, and the assumed transition width. The dominant contribution is the approximately 14% energy-scale uncertainty inherited from the calibration of the SD-750 with the FD.

The exposure uncertainty is approximately 3% and is energy independent, since the exposure is calculated geometrically from the number of active hexagons [3]. The uncertainties associated with the detector response include the energy resolution and the composition dependence of the bias and efficiency. The uncertainties in the calibration parameters determined in Chapter 2 are also included.

These sources are treated as uncorrelated, and their effects on both the flux and the fitted spectral parameters are evaluated using the forward-folding procedure developed in the

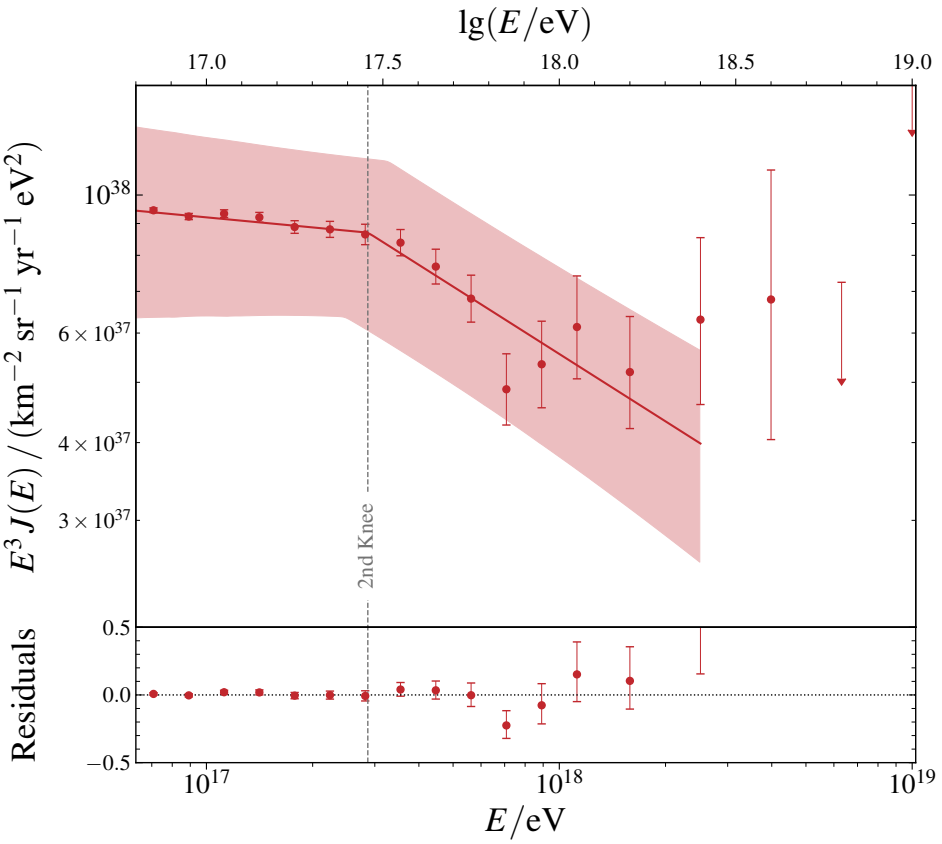


Figure 3.5: Unfolded cosmic-ray spectrum measured with the SD-433 and the best-fit broken power law with a smooth transition. The lower panel shows the residuals with respect to the fitted model.

previous section. Their contributions are combined in quadrature to determine the total systematic uncertainty of the measurement. The influence of the assumed value of ω_{01} on the determination of the second knee is assessed separately using the measured data and toy Monte-Carlo simulations.

3.4.1 FD energy systematic

The calibration of the SD with the FD transfers the systematic uncertainty of the FD energy scale to the SD-433 spectrum. The individual contributions to this uncertainty are described in detail in [25] and summarized in Table 3.2. The dominant contributions arise from the absolute calibration of the FD and the reconstruction of the longitudinal shower profile.

Some contributions exhibit a mild energy dependence, while the total uncertainty is approximately energy independent and amounts to 14%.

The effect of the energy-scale uncertainty on the spectrum is derived analytically. This method contrast with the usual method used in Auger which is based on shifting the energies of measured events. We use this second method as a cross-check of the analytical method we developed.

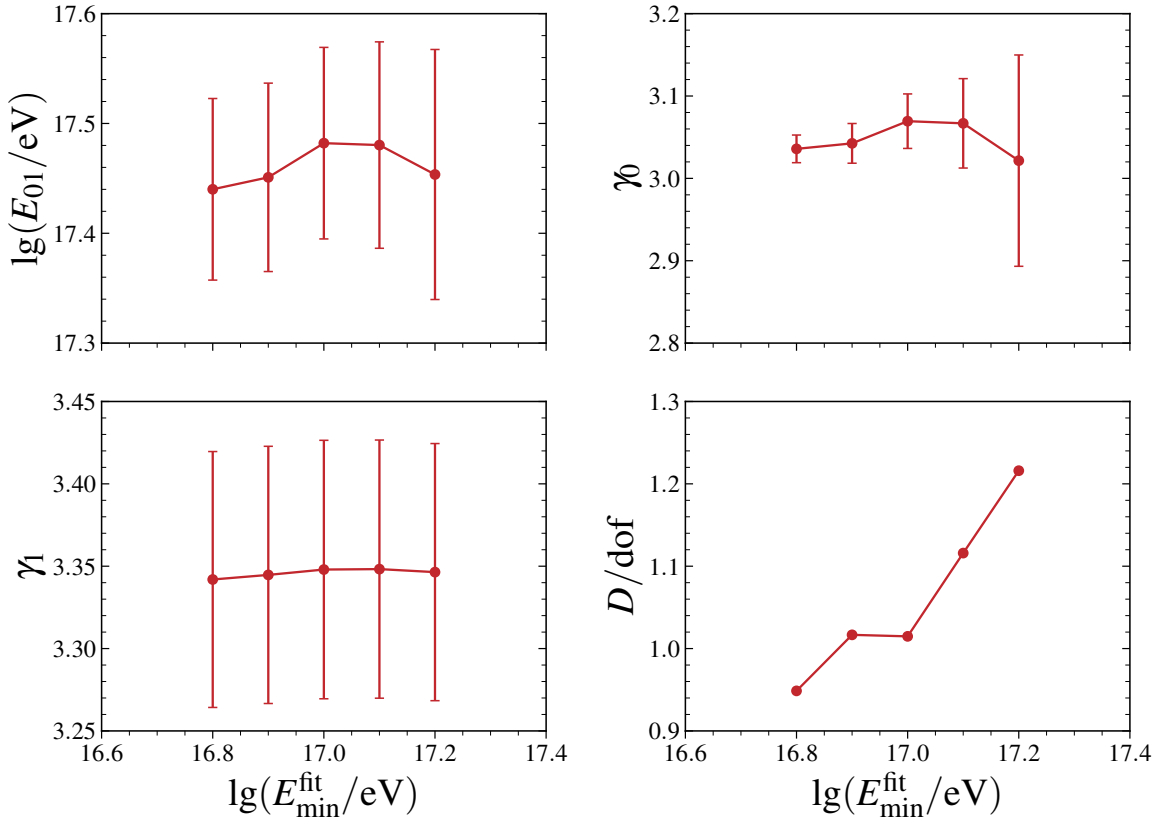


Figure 3.6: Dependence of the fitted second-knee parameters and Poisson deviance per degree of freedom on the minimum energy included in the fit. The error bars represent the statistical uncertainties of the fitted parameters.

Table 3.2: Contributions to the systematic uncertainty in the FD energy scale. The ranges indicate the energy dependence of the corresponding contributions. The total uncertainty amounts to 14 %.

Fluorescence yield	3.6 %
Atmosphere	3.4–6.2 %
FD calibration	9.9 %
FD profile reconstruction	5.6–6.5 %
Invisible energy	3–1.5 %
Statistical error of SD calibration fit	0.7–1.8 %
Stability of the energy scale	5 %
Total	**14 %**

For a relative shift δ in the energy scale, the shifted energy is $E' = (1+\delta)E$, with $\delta = \pm 0.14$. The corresponding change in the flux follows from the transformation of a probability density,

$$J'(E') = J(E) \left| \frac{dE}{dE'} \right|, \tag{3.14}$$

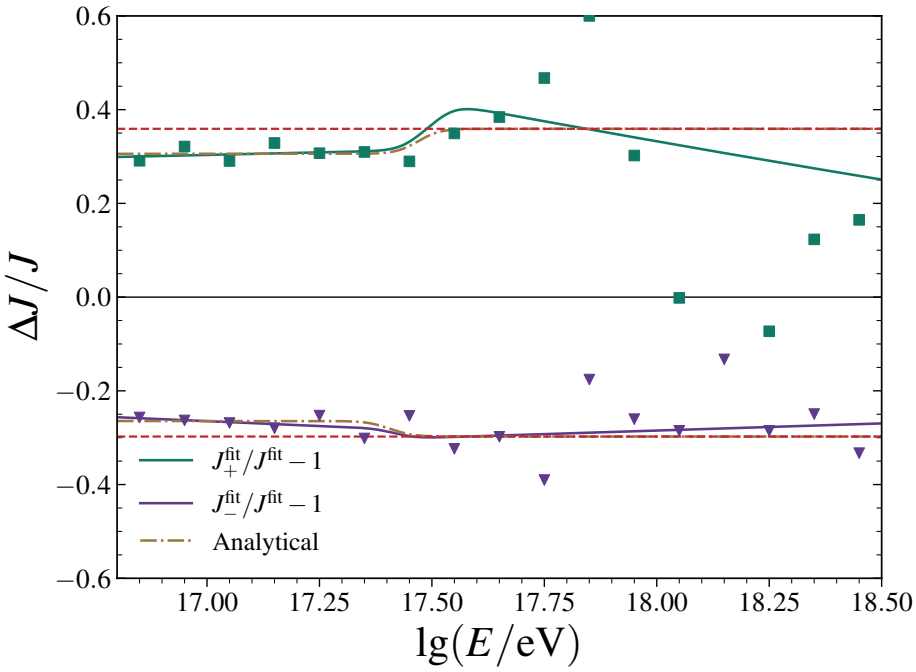


Figure 3.7: Relative change in the fitted flux after coherently shifting the reconstructed energies by $+14\%$ and -14%. The solid curves show the result of repeating the forward-folding fit, while the dash-dotted curves show the analytical expectation from Eq. (3.16). The horizontal dashed lines indicate the corresponding asymptotic simple-power-law expectations.

where $J(E)$ and $J'(E')$ are the differential fluxes on the original and shifted energy scales, respectively. For a power-law spectrum, $J(E) = J_0\, E^{-\gamma}$, the relative change in the flux is,

$$\frac{\Delta J}{J} = (1+\delta)^{\gamma-1} - 1, \tag{3.15}$$

For $\gamma = 3.35$, Eq. (3.15) gives flux shifts of approximately 36 % and -30 % for $\delta = +0.14$ and -0.14, respectively.

For a flux of a general shape, Eq. (3.14) can be expanded to,

$$\frac{\Delta J(E)}{J(E)} = \frac{1}{1+\delta}\frac{J\left(\frac{E}{1+\delta}\right)}{J(E)} - 1, \tag{3.16}$$

The analytical estimate of the systematic uncertainty obtained from the fitted SD-433 spectrum is shown in Fig. 3.7.

As a cross-check, the reconstructed energy of every event is shifted coherently by $\delta = \pm 0.14$, and the complete forward-folding and unfolding procedures are repeated. The migration matrix is kept unchanged because the energy-scale shift does not modify the assumed detector resolution, bias, or efficiency. The corresponding unfolded spectra are shown in Appendix A.

The flux variations obtained with this procedure are also shown in Fig. 3.7. The forward-folding and analytical results agree over most of the energy range. The differences above the second knee arise from the limited statistics above the feature that hinders the estimation of systematics by shifting event energies. Therefore, the analytical method is better suited to accomplish a stable estimation of the systematic uncertainty over the full energy range.

We applied the method of shifting event energies to evaluate the systematic uncertainties of the spectral parameters as shown in Fig. 3.8.

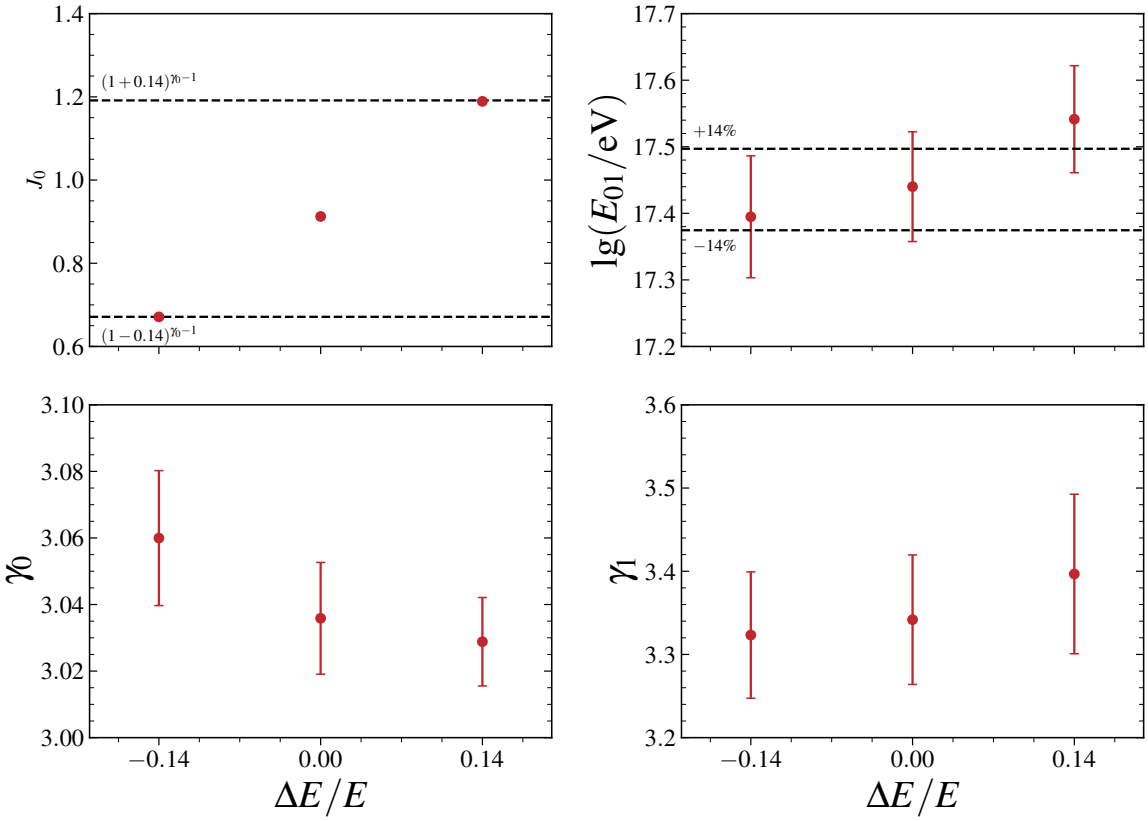


Figure 3.8: Fitted normalization J_0, second-knee energy E_{01}, and spectral indices γ_0 and γ_1 for the nominal energy scale and for coherent shifts of $+14\%$ and -14%. The error bars show the statistical uncertainties of the fits. The dashed lines in the upper panels indicate the expectations obtained by directly propagating the energy-scale shift.

The changes in γ_0 and γ_1 are consistent with statistical uncertainties, in accordance with what is expected, given that a uniform change in energy does not impact the slope of the spectrum. The shifts obtained from the forward-folding procedure are larger than these expectations. Therefore, a conservative systematic uncertainty of $\Delta \lg(E_{01}/\mathrm{eV}) = 0.10$ is assigned, corresponding to the largest variation obtained from the fit.

3.4.2 Response function systematics

The systematic uncertainties associated with the detector response are evaluated by varying independently the energy resolution, bias, and efficiency, within their respective uncertainties. The contribution from the energy resolution is determined by varying the parameters of the resolution model defined in Eq. (2.22), following the procedure in [8]. The parameters R_{sh} and R_0 are shifted up and down by their respective 1σ uncertainties. For each of these two variations, the migration matrix is recalculated while keeping the other components of the detector response unchanged, and the spectrum is fitted again. The relative change in the fitted flux $\Delta J/J$ is calculated for each case and combined in quadrature to get the overall systematic uncertainty due to uncertainty in the energy resolution. The contribution from the energy resolution is shown in the left panel of **??**. Its magnitude is largest near the full-efficiency threshold, where it reaches approximately 1.1%, and decreases with energy. The resulting spectra and fitted parameters after shifting the resolution up and down are reported in Appendix A.

The dependence of the detector response on the assumed mass composition is estimated using the bias and efficiency parametrizations obtained in Section 2.4 for pure proton and pure iron primaries. The contributions of the energy bias and the trigger efficiency are evaluated separately, following the same procedure used for the uncertainty of the energy resolution. The variation of the flux due to the assumed bias leave the position of the second knee and the spectral indices essentially unchanged. However, its contribution to $\Delta J/J$ reaches approximately 4.3% near the full-efficiency threshold. The composition dependence of the efficiency produces a smaller contribution to the flux uncertainty, with a maximum of approximately 1.5%, and shifts the position of the second knee downward by $\Delta \lg(E_{01}/\mathrm{eV}) = -0.03$. The contributions from the energy bias and trigger efficiency are shown in the left panel of **??**.

3.4.3 Calibration systematic

The energy assigned to each SD-433 event is obtained from the calibration of its shower size with the SD-750 energy estimator, described in Section 2.3. In addition to the uncertainty inherited from the FD energy scale, the finite precision of the calibration introduces an independent contribution to the spectrum uncertainty. This contribution arises from the uncertainties in the normalization A and slope B of the calibration relation. A variation of A produces a uniform shift in energy, whereas a variation of B produces an energy-dependent shift and therefore modifies the spectral indices. The effect of the calibration slope can be understood analytically; for the calibration relation, a change from B to B', while keeping A fixed, transforms the reconstructed energy according to,

$$E' = A\left(\frac{E}{A}\right)^{\frac{B'}{B}}. \tag{3.17}$$

The corresponding transformation of the flux follows from the change of variables in Eq. (3.14),

$$J'(E') = \frac{B}{B'}\frac{E}{E'}J(E). \tag{3.18}$$

For a power law spectrum, $J(E) \propto E^{-\gamma}$, the transformed spectrum is also a power law with spectral index,

$$\gamma' = 1 + (\gamma - 1)\frac{B}{B'}. \tag{3.19}$$

This yields a change in the spectral index,

$$\Delta\gamma = (\gamma - 1)\left(\frac{B}{B'} - 1\right). \tag{3.20}$$

In the systematic evaluation, the calibration parameters are varied according to the systematic uncertainties determined in Section 2.3. The correlated variations used to define the systematic envelope are $(A^+, B^-) = (125.7\ \mathrm{PeV}, 0.909)$ and $(A^-, B^+) = (113.3\ \mathrm{PeV}, 0.983)$. The nominal and varied calibration relations are shown in the left panel of Fig. 3.9. For each variation, the energy of every event is recalculated and the spectrum is fitted again. The two resulting unfolded spectra are shown in the right panel of Fig. 3.9. Using the fitted values of the spectral indices, $\gamma_0 = 3.04$ and $\gamma_1 = 3.34$, the analytical method of Eq. (3.20) predicts $\Delta\gamma_0^{\text{analytical}} = 0.083$ and $\Delta\gamma_1^{\text{analytical}} = 0.095$. The corresponding changes obtained from the alternative method of shifting the event energies, are $\Delta\gamma_0^{\text{fit}} = 0.073$ and $\Delta\gamma_1^{\text{fit}} = 0.081$. The

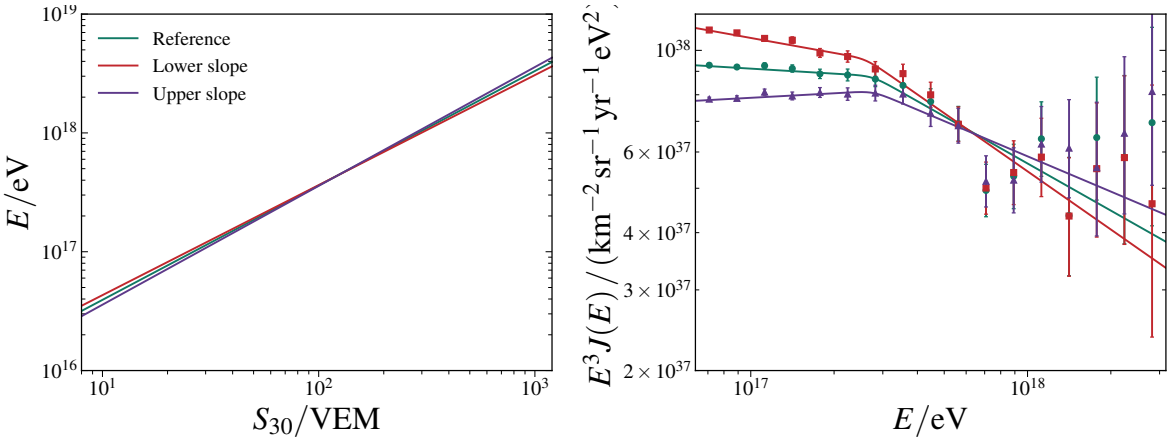


Figure 3.9: Effect of the calibration systematic on the reconstructed spectrum. The left panel shows the reference calibration relation together with the lower- and upper-slope variations. The right panel shows the corresponding unfolded spectra scaled by E^3 together with their respective fitted spectral functions.

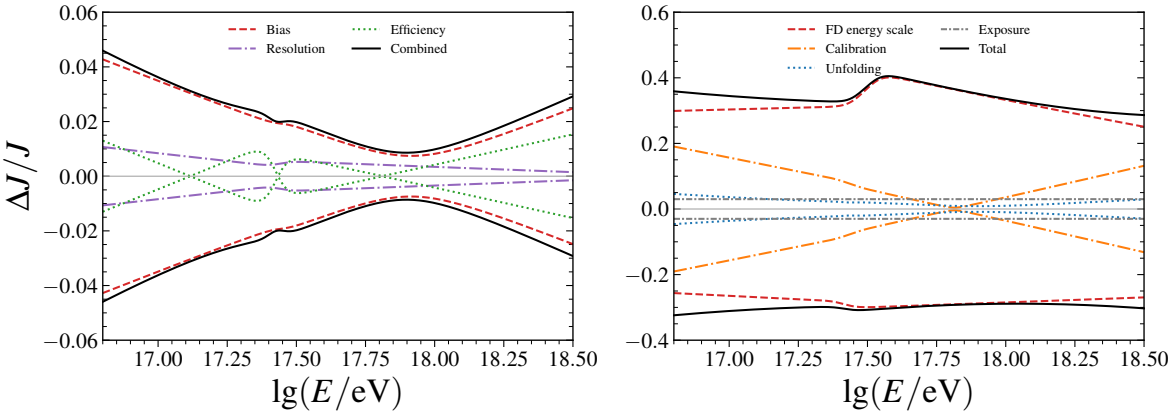


Figure 3.10: Relative systematic uncertainty in the SD-433 cosmic-ray flux. Left: contributions associated with the assumptions used to calculate the migration matrix, obtained by varying the energy bias, efficiency, and resolution. The black curves show their quadratic sum. Right: systematic uncertainty, including the FD energy scale, calibration parameters, combined unfolding contribution, and exposure. The black curves show the upper and lower limits by adding all contributions in quadrature.

systematic uncertainties estimated with both methods agree with each other. Since the analytical estimates are slightly larger, they are adopted as conservative systematic uncertainties. A similar procedure is applied to estimate the contribution of the calibration uncertainty to the position of the second knee of $\Delta \lg(E_{01}/\text{eV}) = 0.024$.

The contribution of the calibration to the flux uncertainty is shown in the right panel of Fig. 3.10. It is the second-largest contribution after the energy-scale uncertainty and its magnitude is largest at the boundaries of the energy range, reaching 19% at $\lg(E/\text{eV}) = 16.8$ and 13% at $\lg(E/\text{eV}) = 18.5$. The right panel of Fig. 3.10 also shows the contributions from the exposure and unfolding assumptions, together with the total uncertainty obtained by combining all contributions in quadrature.

3.4.4 Transition width systematic

As discussed in Section 3.3, leaving the transition width free yields $\omega_{01}^{\text{free}} = 0.092 \pm 0.267$. The large uncertainty, which extends into the unphysical region of negative values,

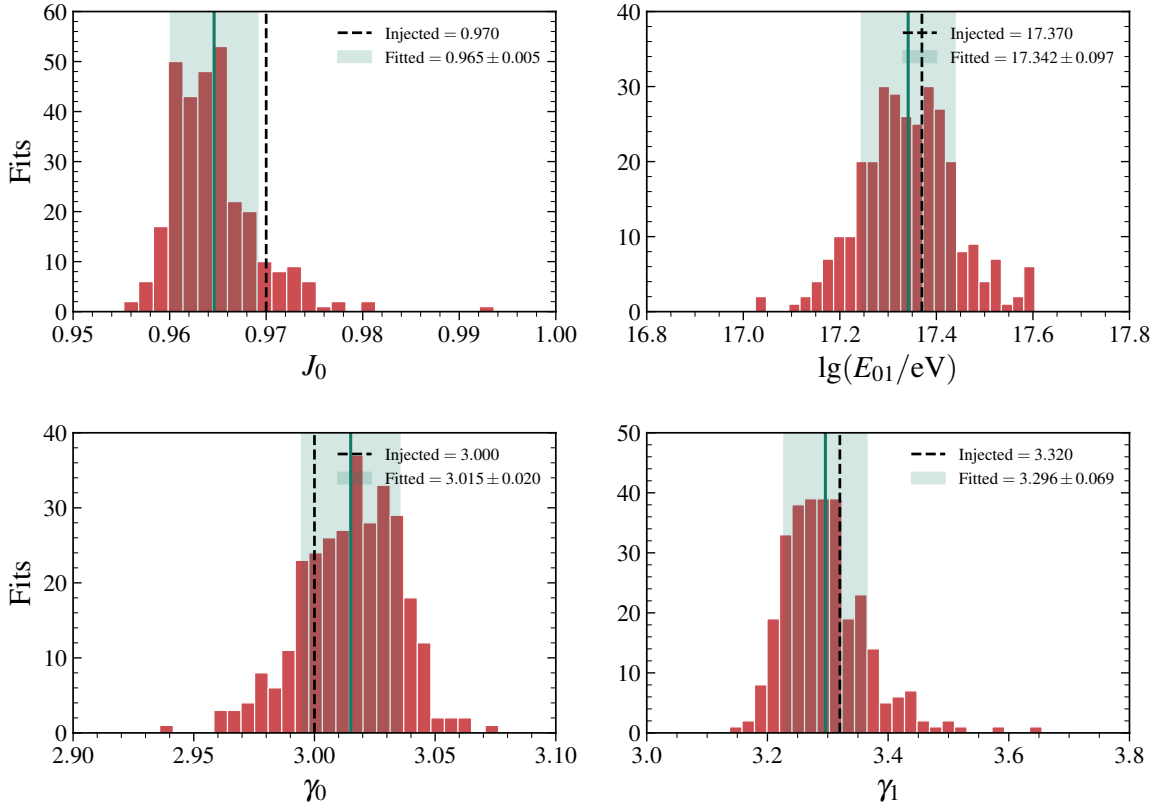


Figure 3.11: Distributions of the spectral parameters reconstructed from toy spectra generated with $\omega_{01} = 0.25$ and fitted with a sharp transition fixed to $\omega_{01} = 0.05$, as used for the measured data. The dashed vertical lines indicate the injected values, while the solid vertical lines show the mean fitted values and the shaded regions represent the corresponding 1σ intervals.

shows that the SD-433 data cannot constrain this parameter. Its impact on the spectrum measurement is therefore studied using toy Monte-Carlo simulations and a scan of the transition width fixed in the fit.

A total of 300 spectra are sampled from Eq. (3.13) using the fitted spectral parameters but with wider transition $\omega_{01} = 0.25$. Each sampled spectra contained 40,000 events, approximately the size of the measured dataset. The detection efficiency is applied during the event sampling, and the energy of each event is smeared according to the response function in Eq. (3.3). Each spectrum is then fitted using the approximate migration matrix from Eq. (3.7).

To test whether fixing the transition width biases the reconstructed spectral parameters, these spectra are fitted with $\omega_{01} = 0.05$. This deliberately compares a broad injected transition with the substantially sharper transition fixed in the fit to the measured data. The distributions of the reconstructed second-knee energy and spectral indices for these spectra are shown in Fig. 3.11, together with their median and injected values. We also estimated the statistical uncertainty of each spectral parameter as the standard deviation of the corresponding histogram. For all cases, the mean of the reconstructed parameter differs from the injected values much less than the statistical uncertainty. This results shows that fixing an incorrect transition width does not introduce a significant bias at the current level of statistics.

To assess the effect of the assumed transition width on the other spectral parameters, the fit is repeated with ω_{01} fixed between 0 and 0.5 in steps of 0.05, with $\omega_{01} = 0$ corresponding to the sharp broken-power-law limit. The resulting Poisson deviance, second-knee energy,

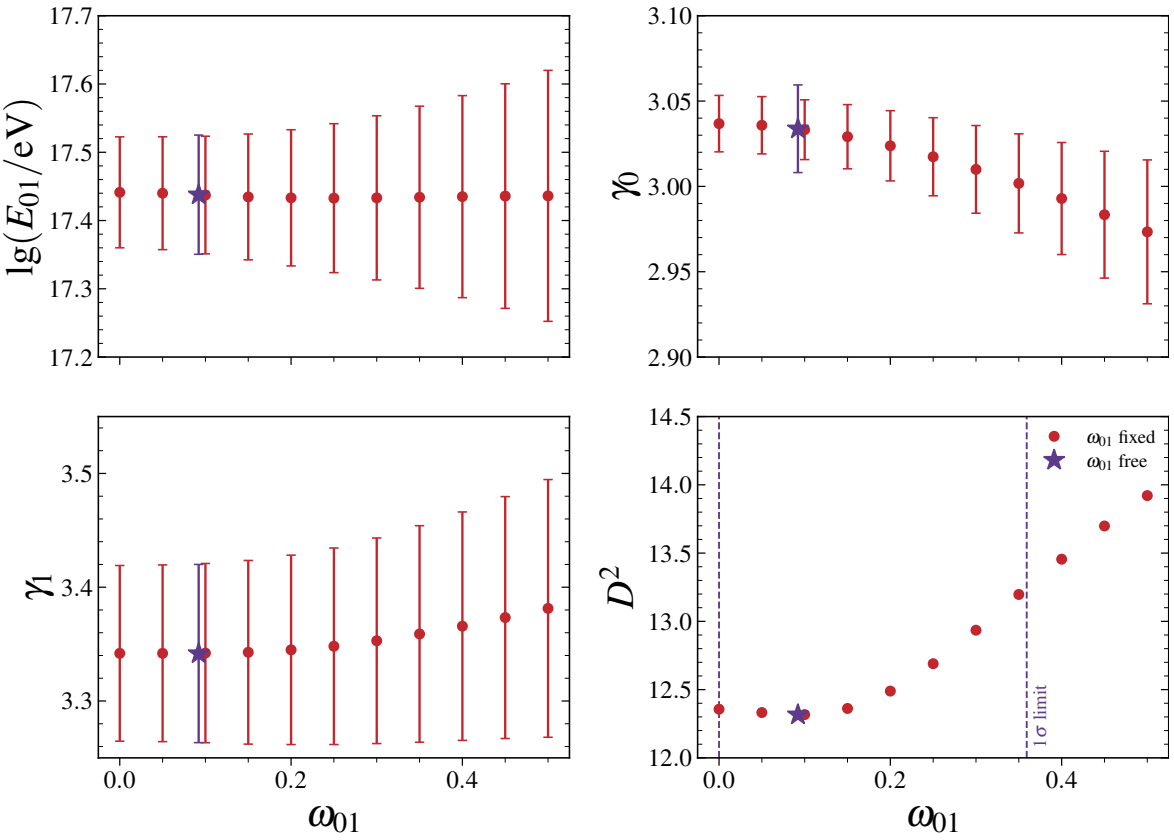


Figure 3.12: Dependence of the fitted second-knee parameters and Poisson deviance on the assumed transition width ω_{01} for the SD-433 spectrum. Circles show fits with ω_{01} fixed, while the star indicates the result obtained when ω_{01} is left free. The error bars represent the statistical uncertainties of the fits. The dashed vertical lines in the lower-right panel indicate the 1σ limits of the free-ω_{01} fit.

and spectral indices are shown in Fig. 3.12. The circles represent the fits with fixed ω_{01} and the stars indicates the result obtained when ω_{01} is left free. The deviance varies by only approximately 1.6 over the complete scan, indicating that the SD-433 data have little sensitivity to the transition width. The reconstructed second-knee energy remains compatible within statistical uncertainties with the nominal result obtained for $\omega_{01} = 0.05$.

A mild correlation is observed between the transition width and the spectral indices. As ω_{01} increases, the fitted value of γ_0 decreases from approximately 3.04 to 2.97, while γ_1 increases from approximately 3.34 to 3.38. This behaviour reflects a compensation between the transition width and the asymptotic slopes: a broader transition is accommodated by a flatter spectrum below the second knee and a steeper spectrum above it. Nevertheless, these variations remain compatible with the statistical uncertainties of the corresponding fits in the neighbourhood of the fitted ω_{01}.

The Monte-Carlo studies and the scan of the transition width highlight that fixing $\omega_{01} = 0.05$ in the fit does not introduce a significant bias in the reconstructed second-knee energy or spectral indices, even if the injected transition is substantially broader. No additional systematic uncertainty is therefore assigned to the choice of ω_{01}.

A summary of the contributions to the systematic uncertainties of the fitted spectral parameters is presented in Table 3.3. The FD energy scale dominates the uncertainties in the normalization and second-knee energy, while the calibration parameters provide the largest contribution to the uncertainties in the spectral indices. The detector-response uncertainties are subdominant throughout the measured energy range and the exposure uncertainty affects

Table 3.3: Contributions to the systematic uncertainties of the fitted SD-433 spectral parameters. The values correspond to the largest absolute variation obtained for each source. The uncertainty in J_0 is expressed in units of $10^{-13}\,\mathrm{km}^{-2}\,\mathrm{yr}^{-1}\,\mathrm{sr}^{-1}\,\mathrm{eV}^{-1}$. The total is obtained by adding the independent contributions in quadrature.

Source	ΔJ_0	$\Delta \lg(E_{01}/\mathrm{eV})$	$\Delta\gamma_0$	$\Delta\gamma_1$
FD energy scale	0.277	0.100	0.024	0.055
Calibration	0.156	0.024	0.083	0.095
Energy resolution	0.008	0.003	0.005	0.002
Composition: bias	0.038	0.006	0.026	0.019
Composition: efficiency	0.005	0.032	0.019	0.011
Exposure	0.028	0	0	0
Total	**0.321**	**0.108**	**0.092**	**0.112**

Table 3.4: Best-fit values of the SD-433 spectral parameters. The first uncertainty is statistical and the second systematic. The transition width is fixed in the fit.

Parameter	Value $\pm\sigma_{\mathrm{stat}} \pm \sigma_{\mathrm{syst}}$
$J_0 / \left(\mathrm{km}^{-2}\,\mathrm{yr}^{-1}\,\mathrm{sr}^{-1}\,\mathrm{eV}^{-1}\right)$	$(0.912 \pm 0.005 \pm 0.321) \times 10^{-13}$
γ_0	$3.04 \pm 0.02 \pm 0.09$
E_{01}/eV	$(2.8 \pm 0.5 \pm 0.7) \times 10^{17}$
γ_1	$3.34 \pm 0.08 \pm 0.11$
ω_{01}	0.05 – fixed

only the normalization. Since the different sources are treated as uncorrelated, the total systematic uncertainty for each parameter is obtained by adding the individual contributions in quadrature.

3.5 Summary of the chapter

This chapter presents the measurement of the cosmic-ray energy spectrum with the SD-433 array using data reconstructed from the previous chapter and recorded between July 2018 and December 2021. The raw flux was obtained from the observed event counts and the geometrical exposure determined in Chapter 1.

To remove detector effects in the spectrum measurement, a forward-folding procedure was developed to be included in the spectrum fit. The detector response was incorporated into a migration matrix describing the probability that an event with true energy in bin j is reconstructed in bin i. The matrix accounts for the energy-dependent detection efficiency, bias, and finite resolution of the SD-433. Although its exact definition depends on the assumed spectral model, this dependence can be neglected when the true-energy binning is sufficiently fine. A true-energy bin width of $\Delta \lg(E_j/\mathrm{eV}) = 0.01$ was therefore adopted, compared with $\Delta \lg(E_i/\mathrm{eV}) = 0.1$ in reconstructed energy.

This approximation was benchmarked before it was applied to the measured spectrum. The expected reconstructed event counts obtained with the approximate matrix were compared with those calculated using the exact, spectrum-dependent expression for an injected power-law spectrum. The relative differences were found to be negligible across the full energy range. This validation showed that the approximate matrix preserves the detector response while avoiding the need to assume the spectral parameters a priori and recalculate the matrix at every step of the fit.

The validated migration matrix was subsequently used in Poisson-likelihood fit to predict the number of events in each bin. Once the best-fit spectral parameters were obtained, the ratio between the expected true and reconstructed event counts provided the correction factors used to unfold the measured flux.

A simple power law model for the flux is rejected with a p-value of 3.1×10^{-4}, providing evidence for a change in the spectral slope. The spectrum is therefore described using a broken power law with a smooth transition. The complete set of fitted parameters is summarized in Table 3.4.

The available SD-433 statistics are insufficient to constrain the transition width, therefore this parameter is fixed to $\omega_{01} = 0.05$, following the value adopted by the Pierre Auger Observatory for spectral features whose transition widths could not be constrained. Our analyses show that this choice does not introduce a significant bias in the reconstructed second-knee energy or spectral indices. The transition width will be investigated with greater sensitivity using the combined SD-433 and SD-750 spectrum, thank to the larger statistics available in the latter.

The systematic study identifies the inherited FD energy-scale uncertainty as the dominant contribution to the flux normalization and the position of the second knee. The systematic uncertainty in the SD-433 energy calibration parameters produces the next-largest contribution to the flux and has the strongest effect on the spectral indices, while uncertainties associated with the detector response lead to smaller contributions. Therefore, the measurement establishes the spectral increase in the energy range covered by SD-433 and measures all related spectral parameters with their corresponding statistical and systematic uncertainties.

Chapter 4

The combined SD spectrum

In this chapter we present an update of the SD-750 spectrum, previously published in 2021 [8] together with the combined SD-433-SD-750 spectrum. For the SD-750 the efficiency measured with hybrid events and simulations from Refs. [8**?**] is revisited and parameterized in a data-driven method using the events in common with the SD-433 in **??**. An update of the dataset is presented in **??** to determine the raw spectrum. The spectrum is then unfolded using the updated response function and unfolding in Section 4.1.3. Finally the two spectrum measured individually with SD-433 and the SD-750 are combined using a likelihood method together with a study of the robustness of the measurement in Section 4.3

4.1 The SD-750 Spectrum

4.1.1 The SD-750 dataset

In 2021, the Pierre Auger Collaboration presented a measurement of a turn-down in the cosmic ray flux at around 10^{17} eV, consisting in about 560 000 events. However, the feature could not be constrained due to the closeness with the minimum energy in the flux measurement, $\lg(E/\mathrm{eV}) = 17$. Therefore, the width of the transition and the spectral index above the feature were fitted from the spectrum. Given the much larger surface of the SD-750, energies up to $\lg(E/\mathrm{eV}) = 19.5$ were used in the spectrum determination. This allowed for the ankle in the cosmic ray spectrum to be measured.

The SD-750 started acquiring Phase I data when including the Gen II triggers, ToTd and MoPS, in Jan 2014 and ended with the inclusion of the new updated electronics on 29 November 2021. The geometrical reconstruction is analogous to the one presented in Section 2.1.1. However, in this case the reconstruction was performed using the other algorithm reconstruction, Herald, given that the Observer reconstruction was not updated for the SD-750.The fiducial 6T5 trigger is also applied to ensure the proper reconstruction of the events and given the inclusion of the Gen. II triggers with a strong sensitivity to the electromagnetic component of the shower, only events with $\theta < 40^{\circ}$ are included in the spectrum measurement. Time periods with acquisition problems are also excluded in the spectrum. When removing the bad periods and using the number of hexagons, analogously to the SD-433, the exposure of the SD-750 can be calculated, obtaining $\mathcal{E}_{\text{SD-750}} = (165.4 \pm 5.0)\ \mathrm{km}^2\,\mathrm{sr\,yr}$above the full efficiency threshold.

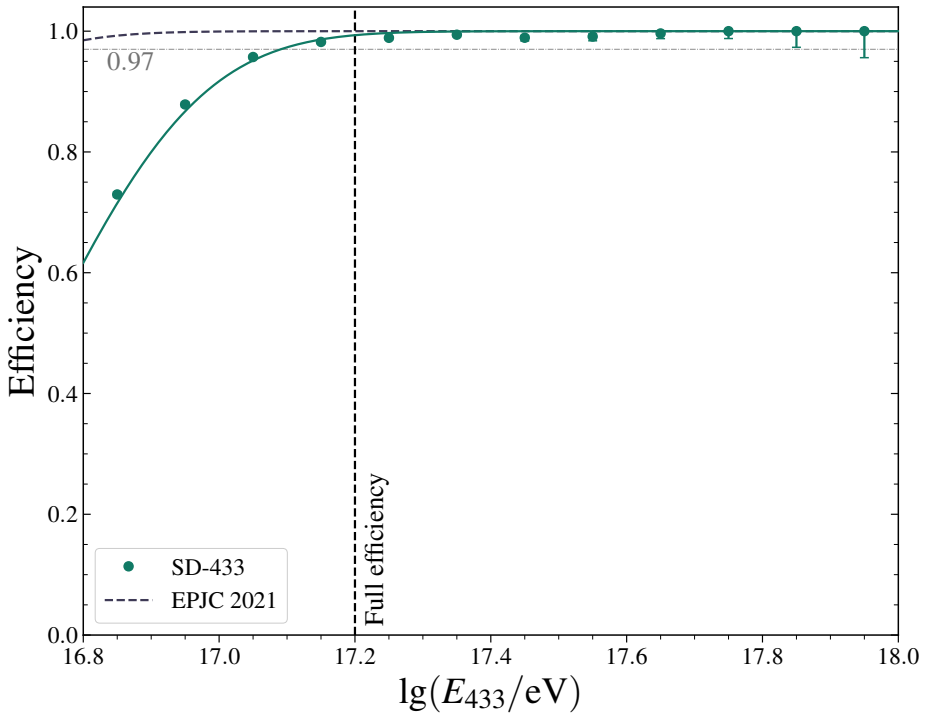


Figure 4.1: Reconstruction efficiency of the SD-750 as a function of the energy reconstructed with the SD-433. The filled markers show the efficiency obtained with the updated dataset, while the open markers show the efficiency reported in the previous SD-750 spectrum measurement [8]. The horizontal dashed line indicates an efficiency of 0.97.

4.1.2 SD-750 efficiency and response function

The SD-750 efficiency was previously estimated using the limited hybrid dataset from events in common with Cohiueco and from simulations [**?**]. Since the SD-433 is nested within the SD-750, opens the possibility to estimate the efficiency from the events in common between the two detectors. Having in mind that there is the caveat that the two energy scales are interwined, hoewever we can study the efficiency since the most important contribution to this study is the full efficiency threshold. The parameterization for the efficiency can also be estimated from this study but the contribution in the spectrum is solely in the forward-folding and hence is minor. Using the events when both detectors were active in phase I, the binned efficiency of the SD-750 can be estimated as

$$\varepsilon(E_i) = \frac{k_i}{n_i}, \tag{4.1}$$

where n_i is the number of events in the i-eth bin detected by the SD-433 and k_i is the number of events detected in common by the SD-750. The estimation was performed, as with the calibration, from the unique id tag of the events and double checked with the GPS time. In Fig. 4.1, the circle data points represent the estimated efficiency of the SD-750 together with the 68% Clopper-Pearson confidence intervals in the error bars [**?**]. The efficiency results is shown for energies above the full efficiency threshold of the SD-433.

This confidence intervals were used to ensure that for $\varepsilon = 1$, the confidence interval reduces to a lower limit, compatible with the physical constrain of the efficiency. To parameterize the efficiency, a binomial likelihood fit was used where for each bin,

$$\mathcal{L}(\mu, \sigma) = \prod_{i}^{N_{bins}} \binom{n_i}{k_i} \varepsilon_i^{k_i} (1 - \varepsilon_i)^{n_i - k_i}. \tag{4.2}$$

The efficiency is modelled using the same parameterization from Ref. [8],

$$\varepsilon = \frac{1}{2} \left[1 + \text{erf} \left(\frac{lg(E/\text{eV}) - \bar{}}{\sigma} \right) \right] \tag{4.3}$$

where $\text{erf}(E)$ is the error function and μ and σ are the shape parameters that govern the transition. The best fit parameters are $\mu = 16.74 \pm 0.01$ and $\sigma = (0.250 \pm 0.001)$ with a correlation $\rho_{\mu,\sigma} = 0.38$. The fit result is shown with a solid line in Fig. 4.1, together with the fit result from Ref. [8] in a dashed line. The gray vertical line shows the full efficiency threshold for the SD-750 estimated with the SD-433. It is noteworthy that the resolution previously estimated reaches full efficiency almost two bins below this work.

In order to construct the migration matrix of the SD-750, the resolution and bias are needed. For this, the model used in Ref. [8] is used. The left panel of Fig. 4.2 shows the relative resolution of the SD-750 from below full efficiency $\lg(E/\text{eV}) = 19.5$, together with the SD-433 resolution At lower energies, the detector contribution of the SD-433 is better. However, as energy increases, the intrinsic detector contribution from SD-750 is lower and the shower-to-shower fluctuations dominate, resulting in a better resolution with this detector. The resolution is also modeled with Eq. (2.22) from **??** with $r_{\text{sh}} = (0.06 \pm 0.02)$ and $r_{\text{det}} = (0.05 \pm 0.01)$. On the right panel of Fig. 4.2, the fit of the relative energy bias is displayed, also for SD-750 and SD-433,

$$b(E) = b_0 \lg(x - b_1) \exp\left[-b_2 (x - b_3)^2\right] + b4 \tag{4.4}$$

with the best fit parameters,

$$b_0 = -3 \pm 1, \qquad b_1 = 26 \pm 4, \qquad b_2 = 0.35 \pm 0.02,$$
$$b_3 = 12.7 \pm 0.1, \qquad b_4 = -0.0039 \pm 0.0008$$

The bias is lower for the SD-433 in the entire energy range. However, for the SD-750 reduces asymptotically to 0.5%.

With the efficiency from the previous section, the SD-750 migration matrix can be constructed with the same formalism from Section 3.3. For this, the approximated matrix with the finer binning in $\Delta E_{\text{true}} = 0.01$ is constructed with Eq. (3.7) to make the matrix independent of the spectrum model. The resulting migration matrix is presented in Fig. 4.3. The dashed vertical line indicates the full efficiency threshold of the SD-750 and the white diagonal line indicates $E_{\text{true}} = E_{\text{recon}}$. As with the SD-433, the matrix is constructed for an energy range wider than the spectrum in order to avoid border effects from the neighbouring bin contributions. Given that the resolution near the SD-750 is similar to the one of the SD-433, the width of the spread in these rows is similar. However, when comparing with Fig. 3.2, at higher energies, the spread around $E_{\text{true}} = E_{\text{recon}}$ is tighter due to the better resolution of the SD-433. With this, using the raw spectrum, the forward folding fit can be performed.

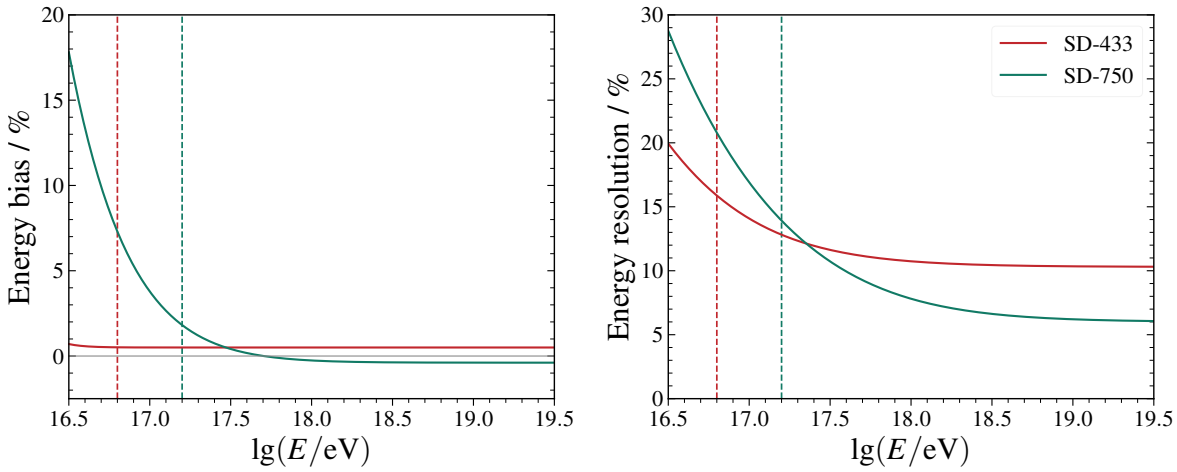


Figure 4.2: Relative energy bias (left) and energy resolution (right) used to construct the SD-433 and SD-750 response matrices. The vertical dashed lines indicate the full-efficiency thresholds adopted for the two arrays.

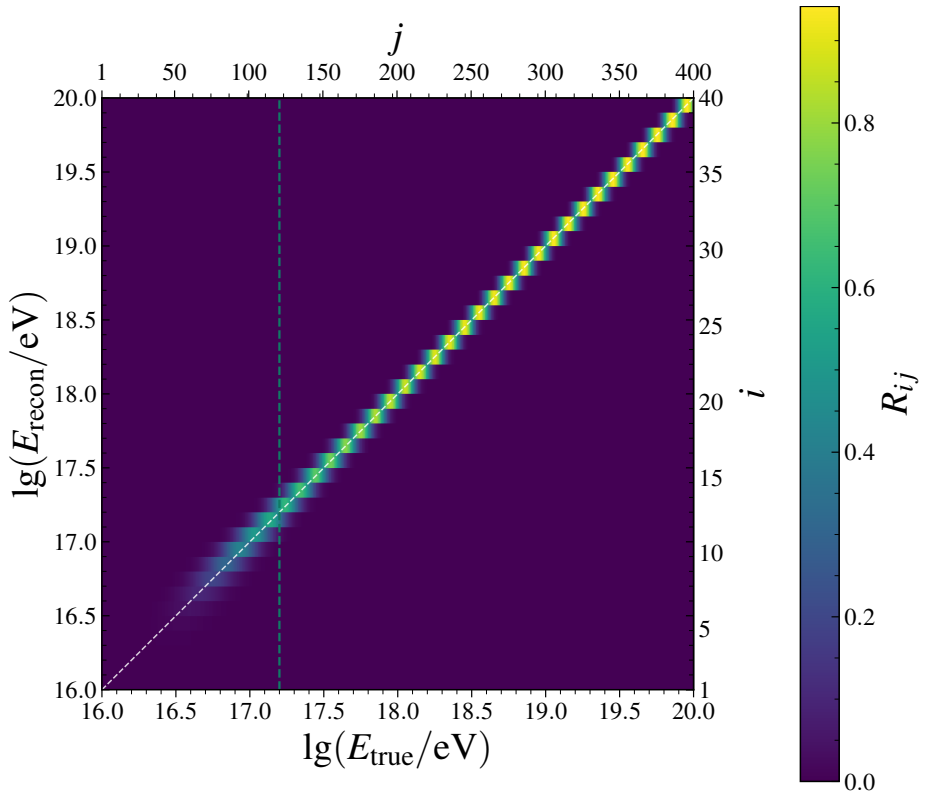


Figure 4.3: Migration matrix used for the updated SD-750 spectrum. The horizontal axis gives the true-energy bin j and the vertical axis the reconstructed-energy bin i; the additional axes give the corresponding logarithmic energies. The matrix includes the measured reconstruction efficiency, energy bias, and energy resolution.

4.1.3 Updated SD-750 spectrum

The SD-750 traditionally uses the Herald reconstruction. For the updated and combined spectrum an update with the ICRC 2025 energy scale was performed as mentioned in the Section 2.3. However, the change in energy is minor with an energy difference less than 0.5% around full efficiency and below 2% at the highest energies. This change in the energy

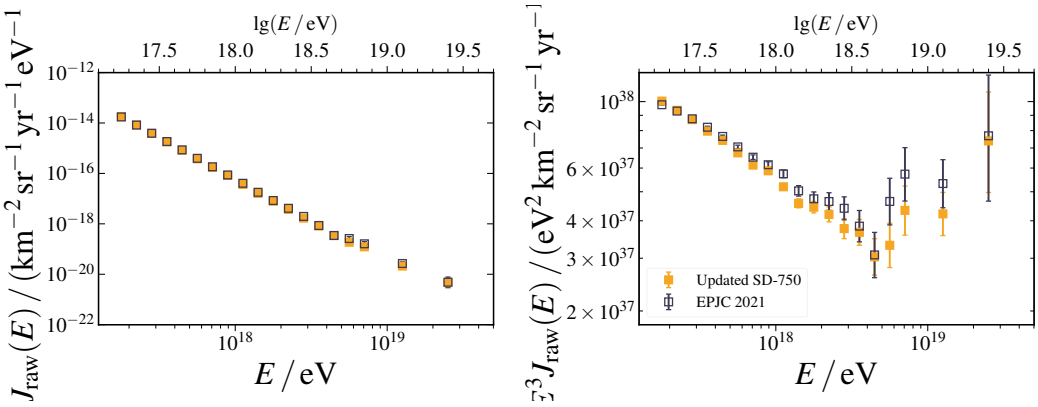


Figure 4.4: Raw cosmic-ray spectrum measured with the updated SD-750 dataset, compared with the spectrum reported in [8]. The left panel shows the differential flux and the right panel the flux multiplied by E^3. Both measurements are displayed using the same energy bins. Only statistical uncertainties are shown.

scale leads to an intrinsic change in the spectrum, however minor. The left panel of Fig. 4.4 shows the SD-750 raw spectrum estimated with the updated dataset for $\theta < 40^\circ$ together with the published in EPJC 2021. The main difference is around 5% at all energy range, due to a bug found in the code where the spectrum used included the bad periods. This difference is constant in energy and can be seen amplified in the right panel of Fig. 4.4. The change in the energy explains the dependence in energy of the raw flux, that increases as the energy increases. However, beyond the normalization, parameters compatible with the EPJC 2021 paper are found when using the same fit conditions, as can be seen in **??**. The updated dataset was used for the SD-433 calibration.

To fit the spectrum, the forward folding using the matrix from Fig. 4.3 was used. The higher energy range of the SD-750 given its larger surface allows to include in the fit the next spectrum feature, the ankle. This feature is a sharp hardening of the spectrum observed at around 4 PeV. To model the flux, an extension of Eq. (3.13) was used,

$$J(E) = J_0 \left(\frac{E}{10^{17}\,\mathrm{eV}}\right)^{\gamma_0} \prod_{i=0}^{1} \left[1 + \left(\frac{E}{E_{ij}}\right)^{\frac{1}{\omega_{ij}}}\right]^{(\gamma_i - \gamma_j)\omega_{ij}}, \tag{4.5}$$

where $j = i + 1$. The parameter E_{12} describes the energy of the ankle, γ_2 is the spectral index after and ω_{12} is the parameter that governs the transition. Given the sharpness of this feature, the transition parameter ω_{12} is fixed to 0.05 during the fit.

When the fit was performed, the second knee energy was fixed to the SD-433 value during the fit, given that the full efficiency threshold of the SD-750 did not allow to constrain all fit the parameters. Nevertheless, when releasing it, $\lg(E_{01}/\mathrm{eV}) = 17.44 \pm 0.16$, very similar to the one obtained from the SD-433 fit. Fig. 4.5 shows the unfolded SD-750 spectrum together with the fit. The resulting fit parameters are shown in Table 4.1. The obtainedγ_1 slope after the second knee is compatible with the measurement from EPJC 2021 and the SD-433 measurement from the previous chapter. The ankle energy and slope are also compatible although the γ_2 is loosely constrained due to the lack of statistics of this array in this energy region. The resulting deviance is $D^2 = 18.07$ for 17 degrees of freedom, corresponding to a p-value of 0.385.

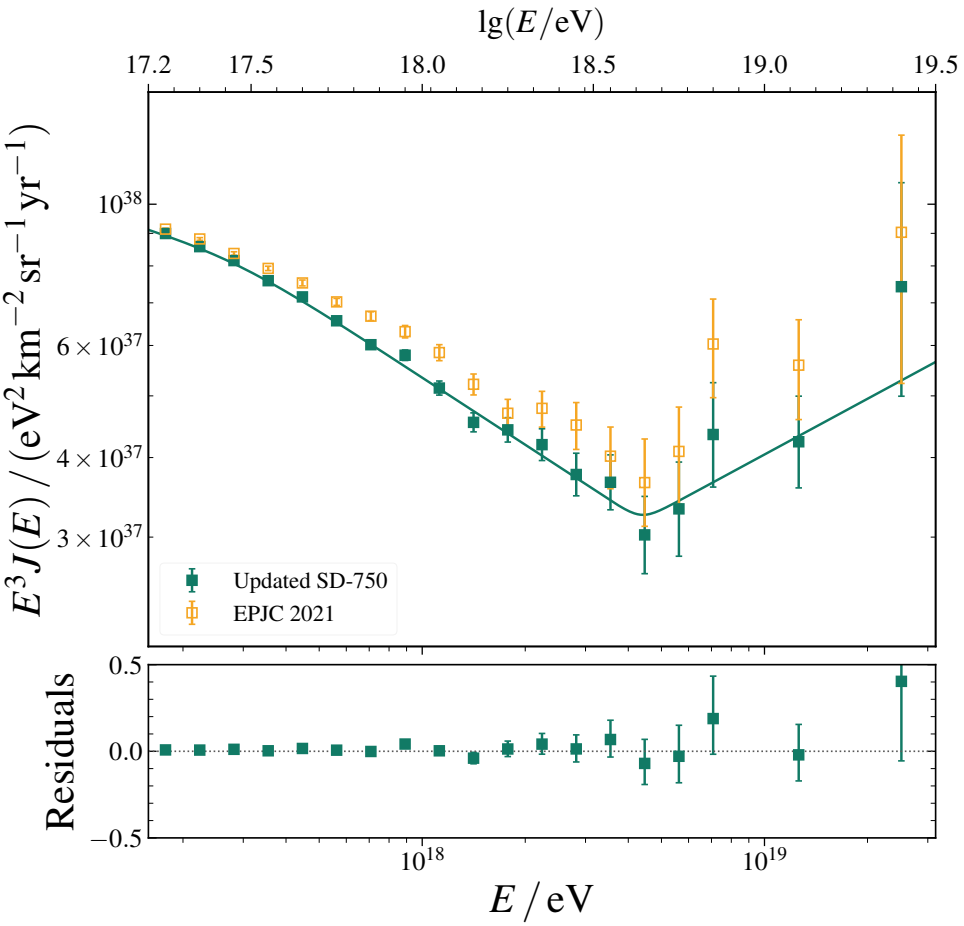


Figure 4.5: Unfolded cosmic-ray spectrum measured with the updated SD-750 dataset and its best-fit spectral model, obtained for $\lg(E/\mathrm{eV}) \geq 17.2$ with the second-knee position fixed to the SD-433 result. The open squares show the unfolded spectrum reported in [8] for comparison. The lower panel shows the residuals of the updated SD-750 points with respect to the fitted model; the published points are not included in the residuals. Only statistical uncertainties are shown.

Table 4.1: Parameters of the standalone SD-750 spectrum fit for $\lg(E/\mathrm{eV}) \geq 17.2$. The second-knee position is fixed to the SD-433 result and the ankle width is fixed to $\omega_{12} = 0.05$. The quoted uncertainties are statistical.

Parameter	Value
J_0 / $(10^{-13}\,\mathrm{km^{-2}\,sr^{-1}\,yr^{-1}\,eV^{-1}})$	0.9861 ± 0.0083
$\lg(E_{01}/\mathrm{eV})$	17.4401 (fixed)
ω_{01}	0.264 ± 0.159
γ_0	3.158 ± 0.030
γ_1	3.351 ± 0.024
$\lg(E_{12}/\mathrm{eV})$	18.641 ± 0.085
ω_{12}	0.05 (fixed)
γ_2	2.709 ± 0.178

4.2 The Combined Spectrum

In order to combine the much higher exposure of the SD-750 with the lower energy threshold of the SD-433 the combined SD-SD spectrum was developed. The approach consist in a combined likelihood as the one in Refs. [8**?**]. Although the two detectors are treated independently, to combine the two spectra, the data sets associated with each detector are treated as independent. In this way, the estimation of the exposure of each detector can be treated individually and in the full efficiency threshold is calculated geometrically as described in section Section 1.4. The combined likelihood is defined as

$$\mathcal{L}(\mathbf{s}) = \mathcal{L}_{\text{SD-433}}\,\mathcal{L}_{\text{SD-750}}\,\mathcal{L}_{\text{exp}}\,\mathcal{L}_{\text{cal}} \tag{4.6}$$

where $\mathcal{L}_{\text{SD-433}}$ and $\mathcal{L}_{\text{SD-750}}$ are the individual likelihoods of the two arrays respectively as defined in Section 3.3. The term $\mathcal{L}_{\text{exp}}$ is a change in the exposure of the SD-433 that allows for the normalization to change. It is modelled as a Gaussian likelihood,

$$\mathcal{L}_{\text{exp}}(\delta\mathcal{E}) = \frac{1}{\sqrt{2\pi}\,\sigma_{\mathcal{E}}} \exp\left[-\frac{1}{2}\left(\frac{\delta\mathcal{E}}{\sigma_{\mathcal{E}}}\right)^2\right], \tag{4.7}$$

where $\delta\mathcal{E}$ is the change in the SD-433 exposure $\mathcal{E}_{\text{SD-433}} - \hat{\mathcal{E}}$, and $\sigma_{\mathcal{E}} = 0.03\,\mathcal{E}_{\text{SD-433}}$ is the 3% relative uncertainty in the exposure [3], penalizing deviations higher than the exposure. An extra change is allowed in the calibration parameters from the SD-433, A and B, since they are measured with the SD-750. In this case, the likelihood is modelled as a two dimensional Gaussian, with σ_A and σ_B the systematic uncertainties determined in Section 2.3 and the correlation ρ_{AB} is obtained from the calibration fit,

$$\mathcal{L}_{AB}(\delta A, \delta B) = \frac{1}{2\pi\sigma_A\sigma_B\sqrt{1-\rho_{AB}^2}} \exp\left[-\frac{1}{2(1-\rho_{AB}^2)}\left(\frac{\delta A^2}{\sigma_A^2} + \frac{\delta B^2}{\sigma_B^2} - \frac{2\rho_{AB}\delta A\,\delta B}{\sigma_A\sigma_B}\right)\right], \tag{4.8}$$

so that the changes in $\delta_A = A - \hat{A}$ and $\delta_B = B - \hat{B}$ are penalized for deviations higher than their systmatic uncertainty.

However, the combined fit must be carried out in two steps because of the degeneracy of $\delta\mathcal{E}$ and δ_A given that by definition, both end up being a proportional factor in the spectrum, as developed in Section 3.4. Because of this a two step fit is performed. In a first step, only the exposure is allowed to change, so the fitted parameters are $\delta\mathcal{E}$ and the free parameters from the spectrum model Eq. (4.5) with exception of ω_{12}. The combined spectrum is shown with purple triangles in Fig. 4.6 together with the SD-433 spectrum in red circles and the SD-750 spectrum in green squares. At lower energies, the information from the SD-433 allows for a proper characterization of the spectral index, γ_1, which is compatible with the SD-433 measurement, while at higher energies, the spectrum is dominated by the higher SD-750 exposure, allowing to characterize the ankle. The measured γ_1 is compatible with the SD-433 and SD-750 measurement, alike the ankle energy and spectral index. However, same as with the SD-433 spectrum, the slope γ_2 is loosely constrained. The exposure-only fit gives $\lg(E_{01}/\text{eV}) = 17.230 \pm 0.058$. The results of this fit and of the subsequent fit with varying calibration parameters are summarized in Table 4.2.

4.3 Combined SD-433 and SD-750 spectrum

4.4 Stability of the combined fit

Table 4.2: Combined SD-433–SD-750 fit results. The shifts $\delta\mathcal{E}_{433}$, δA, and δB are relative to $\mathcal{E}_{433} = 2.888\,\mathrm{km^2\,sr\,yr}$, $A = 149.49$, and $B = 0.963$. In the exposure-only fit, only the exposure shift varies; in the calibration-variation fit, the exposure shift is fixed to the preceding best-fit value and the calibration shifts vary. The ankle width is fixed to $\omega_{12} = 0.05$ in both fits. The quoted errors are those returned by the fits.

Parameter	Exposure only	Calibration variation
J_0 / $(10^{-13}\,\mathrm{km^{-2}\,sr^{-1}\,yr^{-1}\,eV^{-1}})$	0.952 ± 0.039	0.9915 ± 0.0169
$\lg(E_{01}/\mathrm{eV})$	17.230 ± 0.058	17.1719 ± 0.0223
ω_{01}	0.314 ± 0.114	0.3903 ± 0.0385
γ_0	2.938 ± 0.066	2.8682 ± 0.0263
γ_1	3.343 ± 0.018	3.3501 ± 0.0121
$\lg(E_{12}/\mathrm{eV})$	18.605 ± 0.110	18.595 ± 0.113
ω_{12}	0.05 (fixed)	0.05 (fixed)
γ_2	2.882 ± 0.184	2.882 ± 0.186
$\delta\mathcal{E}_{433}$ / $(\mathrm{km^2\,sr\,yr})$	-0.0438 ± 0.0301	-0.04375 (fixed)
δA	0 (fixed)	$+0.04887 \pm 0.00028$
δB	0 (fixed)	-0.003045 ± 0.000002
D^2/dof	36.31/30	29.98/29
p-value	0.198	0.415

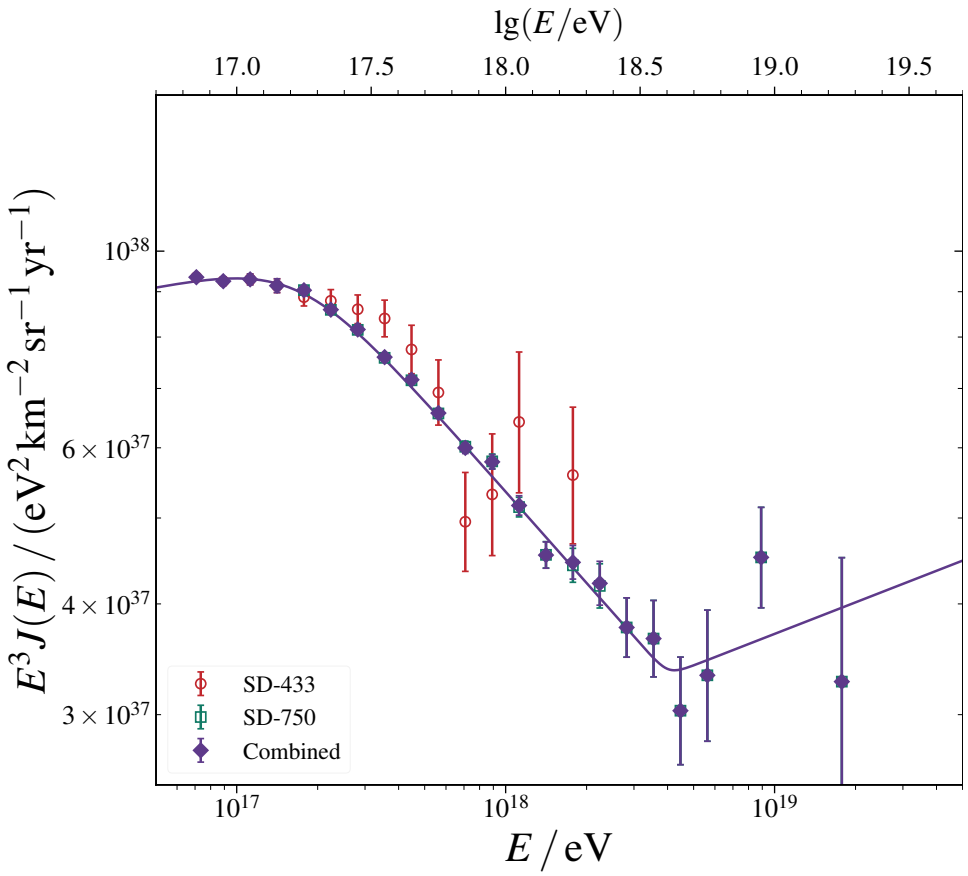


Figure 4.6: Unfolded spectra measured with the SD-433 and SD-750, together with the combined spectrum and common best-fit model. The combined spectrum uses the SD-433 binning below $\lg(E/\mathrm{eV}) = 17.2$ and the SD-750 binning at higher energies. Only statistical uncertainties are shown.

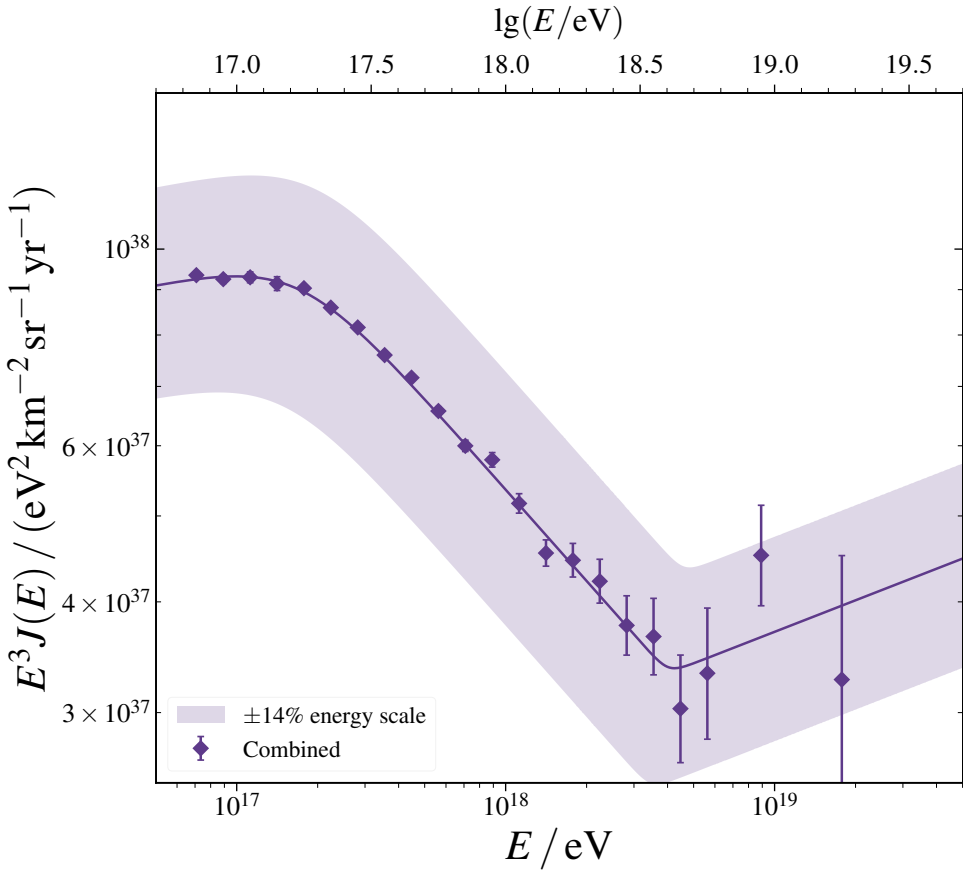


Figure 4.7: Combined SD-433–SD-750 spectrum and common best-fit model. The shaded region shows the effect of the ±14% energy-scale uncertainty propagated through the fitted spectrum using Eq. (3.16). The error bars represent statistical uncertainties.

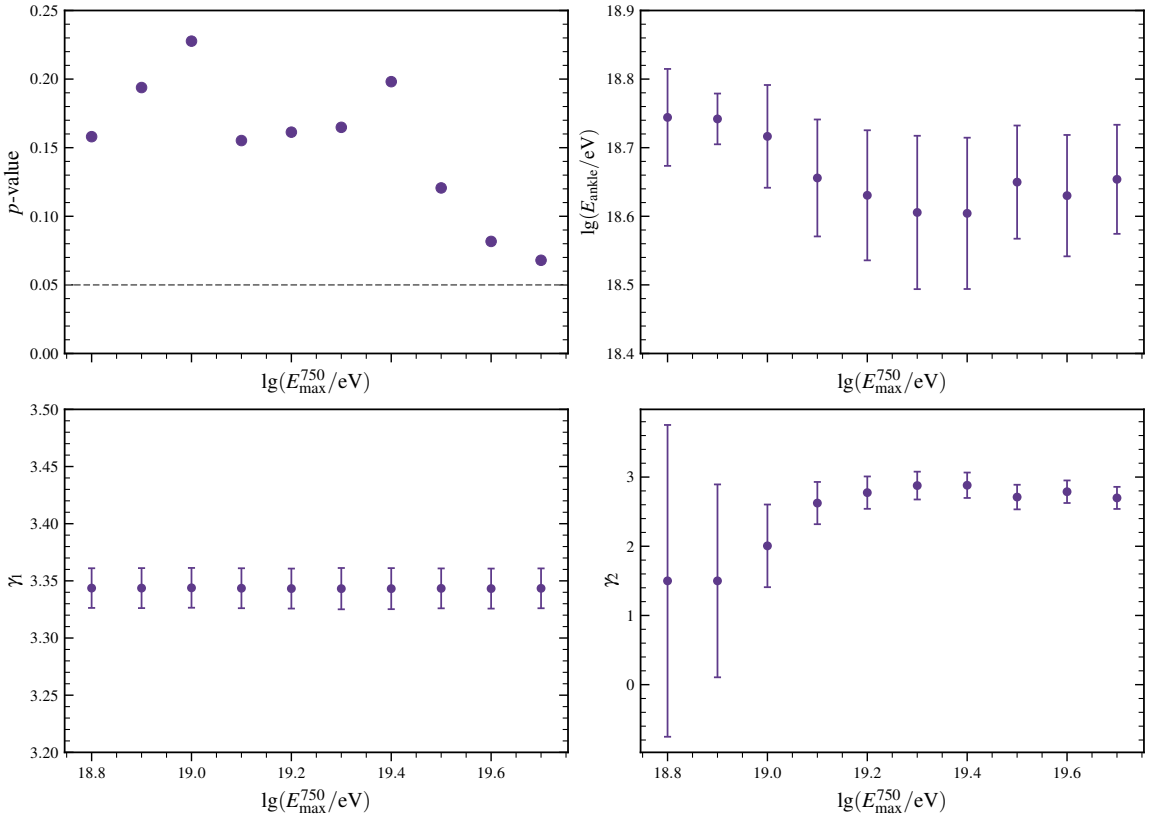


Figure 4.8: Dependence of the combined-fit Poisson deviance, deviance per degree of freedom, and p-value on the maximum SD-750 energy included in the fit. The SD-750 lower boundary is fixed at $\lg(E/\mathrm{eV}) = 17.2$ and the SD-433 fit range is kept unchanged.

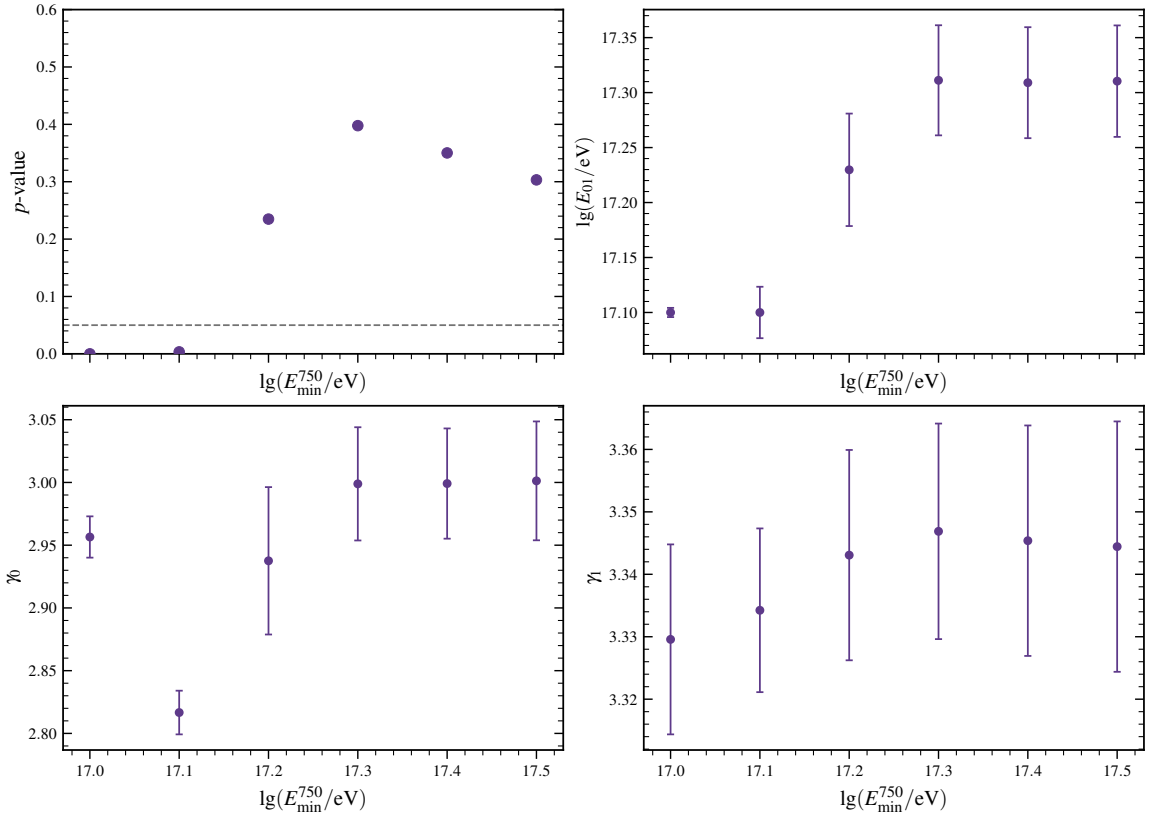


Figure 4.9: Dependence of the combined-fit Poisson deviance, deviance per degree of freedom, and p-value on the minimum SD-750 energy included in the fit. The scan covers $17.0 \leq \lg(E_{\mathrm{min}}/\mathrm{eV}) \leq 17.5$; the fit first becomes acceptable at $\lg(E_{\mathrm{min}}/\mathrm{eV}) = 17.2$. The SD-750 upper boundary is fixed at $\lg(E/\mathrm{eV}) = 19.4$, the SD-433 fit range is unchanged, and the SD-433 exposure is fixed to the value preferred by the nominal combined fit.

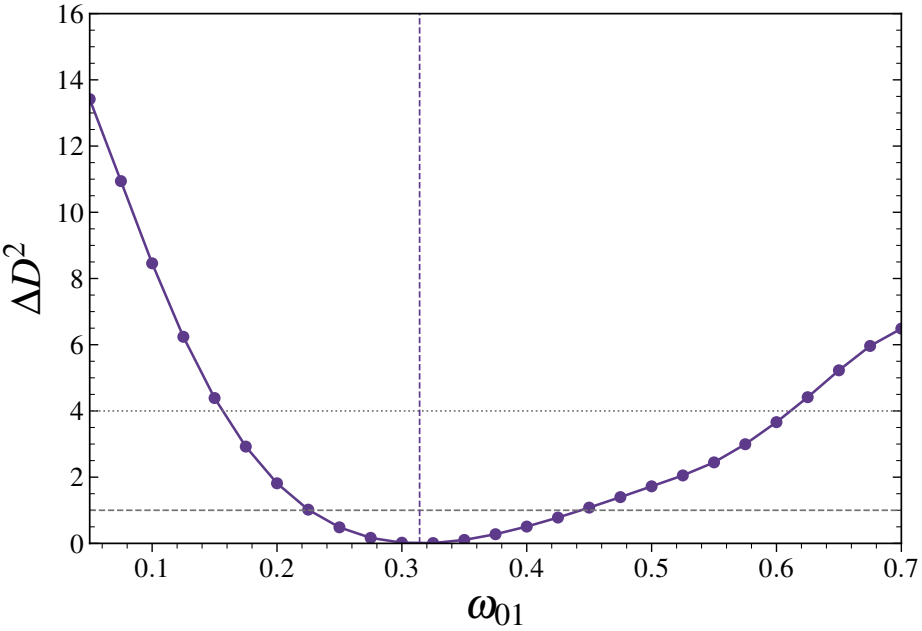


Figure 4.10: Profile of the combined-fit Poisson deviance as a function of the second-knee transition width ω_{01}. At each fixed value of ω_{01}, the remaining spectral parameters are profiled, while the SD-433 exposure is fixed to the value preferred by the nominal combined fit and the calibration parameters are left unchanged. The horizontal lines indicate $\Delta D^2 = 1$ and 4, and the vertical dashed line marks the best-fit value $\omega_{01} = 0.314$.

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Appendix A

Supplementary material for Chapter 3